YEAR 12 MATHEMATICS EXTENSION 1

Chapter 12 Further Calculus

Lesson and Homework Pack

Revision	Trigonometric calculus, inverse trigonometric functions, areas	Advanced · Year 11 Ext 1
Lesson 12A	Inverse trigonometric functions: differentiating	Past papers: 17 marks · 30 min
Lesson 12B	Inverse trigonometric functions: integrating	Past papers: 18 marks · 30 min
Lesson 12C	Further trigonometric integrals	Past papers: 18 marks · 30 min
Lesson 12D	Integration by substitution	Past papers: 19 marks · 30 min
Lesson 12E	Further integration by substitution	Past papers: 20 marks · 30 min
Lesson 12F	Volumes of rotation	Past papers: 20 marks · 30 min
Review	Chapter 12 review exercise	Mixed practice

Each lesson runs in the same order: **concept** → **tutor demo** → **your attempt** → **worked example** to keep for homework. Then do the **textbook homework** and finish with the **past-paper set** under exam conditions (about 30 minutes). Answers and full past-paper solutions are in the separate Homework Answers booklet.

CHAPTER REVISION

What this chapter builds on

Trigonometric calculus (Advanced Chapter 7), the inverse trigonometric functions (Year 11 Extension 1 Chapter 17), and areas by integration (Advanced Chapter 5).

SUMMARY · keep this page open while you work

TRIGONOMETRIC CALCULUS

$$\frac{d}{dx}\sin x = \cos x \qquad \frac{d}{dx}\cos x = -\sin x \qquad \frac{d}{dx}\tan x = \sec^2 x$$

$$\int \cos(ax+b)\,dx = \tfrac{1}{a}\sin(ax+b) + C \qquad \int \sin(ax+b)\,dx = -\tfrac{1}{a}\cos(ax+b) + C$$

$$\int \sec^2(ax+b)\,dx = \tfrac{1}{a}\tan(ax+b) + C$$

THE INVERSE TRIGONOMETRIC FUNCTIONS

Function	Domain	Range	Symmetry
$y = \sin^{-1} x$	$-1 \le x \le 1$	$-\frac{\pi}{2} \le y \le \frac{\pi}{2}$	odd
$y = \cos^{-1} x$	$-1 \le x \le 1$	$0 \le y \le \pi$	point symmetry in $(0, \frac{\pi}{2})$
$y = \tan^{-1} x$	all real x	$-\frac{\pi}{2} < y < \frac{\pi}{2}$	odd

$\sin^{-1}(-x) = -\sin^{-1} x$, $\cos^{-1}(-x) = \pi - \cos^{-1} x$, $\tan^{-1}(-x) = -\tan^{-1} x$.

AREAS AND THE REVERSE CHAIN RULE

Area between a curve and the x-axis: $A = \int_a^b y\,dx$ (take absolute values of parts below the axis).

$$\int f'(x)\,(f(x))^n\,dx = \frac{(f(x))^{n+1}}{n+1} + C \qquad \int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C \qquad \int f'(x)e^{f(x)}\,dx = e^{f(x)} + C$$

WARM-UP · answers are in the answers booklet

1 Differentiate:

a $\sin 3x$ b $x^2 \cos x$ c $\tan(x^2)$

2 Find:

a $\int \cos 2x\,dx$ b $\int_0^{\frac{\pi}{4}} \sec^2 x\,dx$

3 Find the exact values of:

a $\sin^{-1}\frac{\sqrt{3}}{2}$ b $\cos^{-1}\left(-\frac{1}{2}\right)$ c $\tan^{-1}(-1)$

4 Sketch $y = \sin^{-1} x$, $y = \cos^{-1} x$ and $y = \tan^{-1} x$ on separate axes, showing the domain, range and any asymptotes.

5 Use the reverse chain rule to find $\int 2x(x^2+1)^3\,dx$.

6 Find the area between $y = x^2$ and the x-axis from $x = 0$ to $x = 3$.

LESSON 12A

Inverse trigonometric functions: differentiating

Derive and use the derivatives of $\sin^{-1} x$, $\cos^{-1} x$ and $\tan^{-1} x$, with linear and chain-rule extensions, and apply them to tangents, constant functions and curve sketching.

CONCEPT 1 · the three standard derivatives

DERIVING THE DERIVATIVES

Change to the inverse function and use the known derivative of sine, cosine or tangent. Keep track of the restriction on y so that the correct square root (positive or negative) is chosen.

A Let $y = \sin^{-1} x$. Then $x = \sin y$, where $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$.

$$\frac{dx}{dy} = \cos y$$

Because y is in quadrant 1 or 4, $\cos y \ge 0$, so $\cos y = +\sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$. Hence

$$\frac{d}{dx}\sin^{-1} x = \frac{1}{\sqrt{1 - x^2}}.$$

B Let $y = \cos^{-1} x$. Then $x = \cos y$, where $0 \le y \le \pi$.

$$\frac{dx}{dy} = -\sin y$$

Because y is in quadrant 1 or 2, $\sin y \ge 0$, so $\sin y = +\sqrt{1 - \cos^2 y} = \sqrt{1 - x^2}$. Hence

$$\frac{d}{dx}\cos^{-1} x = -\frac{1}{\sqrt{1 - x^2}}.$$

C Let $y = \tan^{-1} x$, so $x = \tan y$, where $-\frac{\pi}{2} < y < \frac{\pi}{2}$. Then $\frac{dx}{dy} = \sec^2 y = 1 + \tan^2 y = 1 + x^2$, and no choice of square root arises:

$$\frac{d}{dx}\tan^{-1} x = \frac{1}{1 + x^2}.$$

STANDARD FORMS, WITH LINEAR EXTENSIONS

$$\frac{d}{dx}\sin^{-1} x = \frac{1}{\sqrt{1 - x^2}} \qquad \frac{d}{dx}\cos^{-1} x = -\frac{1}{\sqrt{1 - x^2}} \qquad \frac{d}{dx}\tan^{-1} x = \frac{1}{1 + x^2}$$

$$\frac{d}{dx}\sin^{-1}(ax + b) = \frac{a}{\sqrt{1 - (ax + b)^2}} \qquad \frac{d}{dx}\cos^{-1}(ax + b) = -\frac{a}{\sqrt{1 - (ax + b)^2}} \qquad \frac{d}{dx}\tan^{-1}(ax + b) = \frac{a}{1 + (ax + b)^2}$$

The derivatives of $\sin^{-1} x$ and $\cos^{-1} x$ only exist for $-1 < x < 1$: the tangents at $x = \pm 1$ are vertical.

TUTOR DEMO **Product rule and a linear extension**

Differentiate: **a** $y = x \sin^{-1} x$ **b** $y = \cos^{-1}(3x - 1)$

YOUR TURN **Same skills**

Differentiate: **a** $y = x^2 \tan^{-1} x$ **b** $y = \tan^{-1}(2x + 1)$

WORKED EXAMPLE 1 **Keep for homework**

Differentiate: **a** $y = x\tan^{-1}x$ **b** $y = \sin^{-1}(ax+b)$

SOLUTION

a Product rule with $u = x$, $v = \tan^{-1}x$, so $u' = 1$, $v' = \dfrac{1}{1+x^2}$:

$$\begin{aligned} y' &= vu' + uv' \\ &= \tan^{-1}x \times 1 + x \times \frac{1}{1+x^2} \\ &= \tan^{-1}x + \frac{x}{1+x^2} \end{aligned}$$

b Chain rule with $u = ax+b$, $y = \sin^{-1}u$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= \frac{1}{\sqrt{1-(ax+b)^2}} \times a \\ &= \frac{a}{\sqrt{1-(ax+b)^2}} \end{aligned}$$

CONCEPT 2 · gradients and tangents

USING THE DERIVATIVE

Set y' equal to the required gradient and solve, or substitute x into y' to find the gradient of a tangent. Always check that the x-values found lie in the domain, and that the y-values are in the range of the inverse function.

TUTOR DEMO **Points with a given gradient**

Find the points on the curve $y = \sin^{-1}(x+1)$ where the tangent has gradient 2.

YOUR TURN **Equation of a tangent**

Find the equation of the tangent to $y = \tan^{-1}x$ at the point where $x = 1$.

WORKED EXAMPLE 2 **Keep for homework**

a Find the points A and B on the curve $y = \cos^{-1}(x-1)$ where the tangent has gradient -2. **b** Sketch the curve, showing these points.

SOLUTION

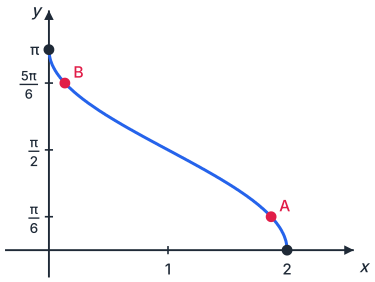
$$y' = -\frac{1}{\sqrt{1-(x-1)^2}} = -2$$

$$1-(x-1)^2 = \tfrac{1}{4}$$

$$(x-1)^2 = \tfrac{3}{4}$$

$$x = 1 + \tfrac{1}{2}\sqrt{3} \text{ or } 1 - \tfrac{1}{2}\sqrt{3}$$

so the points are $A\left(1+\frac{1}{2}\sqrt{3}, \frac{\pi}{6}\right)$ and $B\left(1-\frac{1}{2}\sqrt{3}, \frac{5\pi}{6}\right)$.



CONCEPT 3 · functions whose derivative is zero are constant

PROVING IDENTITIES WITH CALCULUS

If $f'(x) = 0$ for all x in an interval, then $f(x)$ is constant on that interval. To find the constant, substitute any convenient value of x in the interval.

If the domain is split into separate intervals, the constant may be different on each interval.

TUTOR DEMO Show a function is constant

Let $f(x) = \tan^{-1} x - \tan^{-1}\left(\dfrac{x-1}{x+1}\right)$, for $x > -1$. Show that $f'(x) = 0$, and hence find the value of the constant $f(x)$.

YOUR TURN Two intervals

Show that $\dfrac{d}{dx}\left(\tan^{-1} x + \tan^{-1}\dfrac{1}{x}\right) = 0$. Hence find the value of $\tan^{-1} x + \tan^{-1}\dfrac{1}{x}$ for $x > 0$ and for $x < 0$.

WORKED EXAMPLE 3 Keep for homework

a Differentiate $\sin^{-1} x + \cos^{-1} x$. **b** Hence prove the identity $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

SOLUTION

a $\dfrac{d}{dx}\left(\sin^{-1} x + \cos^{-1} x\right) = \dfrac{1}{\sqrt{1-x^2}} + \dfrac{-1}{\sqrt{1-x^2}} = 0$

b Hence $\sin^{-1} x + \cos^{-1} x = C$, for some constant C. Substitute $x = 0$: $0 + \frac{\pi}{2} = C$, so $C = \frac{\pi}{2}$, and $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, as required.

CONCEPT 4 · curve sketching using calculus

SKETCHING WITH THE DERIVATIVE

The usual methods of curve sketching (domain, symmetry, intercepts, a table of values, the sign of y') extend to curves involving inverse trigonometric functions. Watch for square roots such as $\sqrt{\sin^2 x} = |\sin x|$, which give different formulae in different quadrants.

TUTOR DEMO **A sawtooth graph**

Use calculus to sketch $y = \sin^{-1}(\sin x)$ for $-\pi \leq x \leq 2\pi$.

WORKED EXAMPLE 4 **Keep for homework**

Use calculus to sketch $y = \cos^{-1}(\cos x)$.

SOLUTION

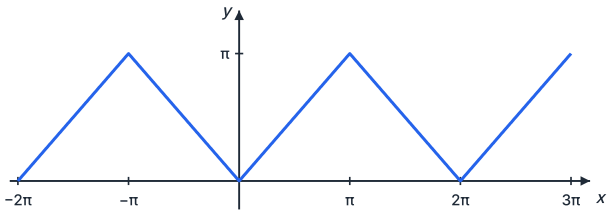
The function is periodic with the same period 2π as $\cos x$. A table of test values gives key points:

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π	$\frac{5\pi}{2}$	3π
y	0	$\frac{\pi}{2}$	π	$\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π

Differentiating with the chain rule ($u = \cos x$, $y = \cos^{-1} u$):

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1-\cos^2 x}} \times (-\sin x) = \frac{\sin x}{\sqrt{\sin^2 x}}$$

When $\sin x > 0$, $\sqrt{\sin^2 x} = \sin x$, so $\frac{dy}{dx} = 1$. When $\sin x < 0$, $\sqrt{\sin^2 x} = -\sin x$, so $\frac{dy}{dx} = -1$. So $y' = 1$ for x in quadrants 1 and 2, and $y' = -1$ in quadrants 3 and 4: the graph is a series of line segments of gradient ± 1.



CONCEPT 5 · chain-rule extensions

CHAIN-RULE EXTENSIONS TO THE STANDARD FORMS

$$\frac{d}{dx}\sin^{-1} f(x) = \frac{f'(x)}{\sqrt{1-(f(x))^2}} \qquad \frac{d}{dx}\cos^{-1} f(x) = -\frac{f'(x)}{\sqrt{1-(f(x))^2}} \qquad \frac{d}{dx}\tan^{-1} f(x) = \frac{f'(x)}{1+(f(x))^2}$$

Equivalently, with $u = f(x)$: $\frac{u'}{\sqrt{1-u^2}}$, $-\frac{u'}{\sqrt{1-u^2}}$ and $\frac{u'}{1+u^2}$.

TUTOR DEMO **Chain-rule formulae**

Differentiate: **a** $\tan^{-1}(e^{2x})$ **b** $\sin^{-1}(x^2)$

YOUR TURN **Same formulae**

Differentiate: **a** $\cos^{-1}(x^3)$ **b** $\tan^{-1}(x^2+1)$

WORKED EXAMPLE 5 **Keep for homework**

Use the chain-rule extension formulae to differentiate: **a** $\tan^{-1} e^{-5x}$ **b** $\cos^{-1}(3x^2)$

SOLUTION

a Here $f(x) = e^{-5x}$ and $f'(x) = -5e^{-5x}$, so

$$\frac{d}{dx}\left(\tan^{-1} e^{-5x}\right) = \frac{f'(x)}{1+\left(f(x)\right)^2} = \frac{-5e^{-5x}}{1+e^{-10x}}.$$

b Here $f(x) = 3x^2$ and $f'(x) = 6x$, so

$$\frac{d}{dx}\left(\cos^{-1} 3x^2\right) = -\frac{f'(x)}{\sqrt{1-\left(f(x)\right)^2}} = -\frac{6x}{\sqrt{1-9x^4}}.$$

TEXTBOOK HOMEWORK · Exercise 12A

FOUNDATION

1 Differentiate with respect to x:

a $\cos^{-1} x$ b $\sin^{-1} 2x$ c $\cos^{-1} 5x$

d $\sin^{-1} x^2$ e $\tan^{-1}(x+2)$ f $x \sin^{-1} x$

g $(1+x^2)\tan^{-1} x$ h $\cos^{-1}\sqrt{x}$ i $\tan^{-1}\dfrac{1}{x}$

2 Find the gradient of the tangent to each curve at the point indicated:

a $y = 2\tan^{-1} x$, at $x = 0$ b $y = \sqrt{3}\sin^{-1} x$, at $x = \frac{1}{2}$

c $y = \tan^{-1} 2x$, at $x = -\frac{1}{2}$ d $y = \cos^{-1}\frac{x}{2}$, at $x = \sqrt{3}$

3 Find, in the form $y = mx + b$, the equation of the tangent and the normal to each curve at the point indicated:

a $y = 2\cos^{-1} 3x$, at $x = 0$ b $y = \sin^{-1}\frac{x}{2}$, at $x = \sqrt{2}$

4 Use the method of Worked Example 3 to show that each function is a constant function, and find the value of the constant:

a $\cos^{-1} x + \cos^{-1}(-x)$ b $2\sin^{-1}\sqrt{x} - \sin^{-1}(2x-1)$

DEVELOPMENT

5 a If $f(x) = x\tan^{-1}x - \frac{1}{2}\ln(1+x^2)$, show that $f''(x) = \dfrac{1}{1+x^2}$.

b Is the graph of $y = f(x)$ concave up or concave down at $x = -1$?

6 Find the derivative of each function in simplest form:

a $x\cos^{-1}x - \sqrt{1-x^2}$

b $\sin^{-1}e^{3x}$

c $\sin^{-1}\frac{1}{4}(2x-3)$

d $\ln\sqrt{\sin^{-1}x}$

e $\sqrt{x}\,\sin^{-1}\sqrt{1-x}$

7 Consider the function $y = \sin^{-1}2x$.

a Write down the range of the function.

b Make x the subject of the equation.

c Find $\dfrac{dx}{dy}$, and explain why it is never negative.

d Use the result that $\dfrac{dy}{dx}$ is the reciprocal of $\dfrac{dx}{dy}$ to find $\dfrac{dy}{dx}$.

8 Consider the function $f(x) = \cos^{-1}x^2$.

a What is the domain of $f(x)$?

b About which line is the graph of $y = f(x)$ symmetric?

c Find $f'(x)$.

d Show that $y = f(x)$ has a maximum turning point at $x = 0$.

e Show that $f'(x)$ is undefined at the endpoints of the domain. What is the geometrical significance of this?

f Sketch the graph of $y = f(x)$.

9 A picture TB that is 1 metre tall is hung on a wall so that its bottom edge B is 3 metres above the eye E of a viewer. Let the distance EP from the viewer to the wall be x metres, and let θ be the angle that the picture subtends at E (P is the point on the wall level with E).

a Show that $\theta = \tan^{-1}\dfrac{4}{x} - \tan^{-1}\dfrac{3}{x}$.

b Show that θ is maximised when the viewer is $2\sqrt{3}$ metres from the wall.

c Show that the maximum angle subtended by the picture at E is $\tan^{-1}\dfrac{\sqrt{3}}{12}$.

10 A plane P at a constant altitude of 6 km and at a constant speed of 600 km/h is flying directly away from an observer at O on the ground. The point A on the path of the plane lies directly above O. Let the distance AP be x km, and let the angle of elevation of the plane from the observer be θ.

a Show that $\theta = \tan^{-1}\dfrac{6}{x}$.

b Show that $\dfrac{d\theta}{dt} = \dfrac{-3600}{x^2+36}$ radians per hour.

c Hence find, in radians per second, the rate at which θ is decreasing at the instant when the distance AP is 3 km.

11 Consider the function $f(x) = \cos^{-1}\dfrac{1}{x}$.

a State the domain of $f(x)$. (Think about it rather than relying on algebra.)

b Recalling that $\sqrt{x^2} = |x|$, show that $f'(x) = \dfrac{1}{|x|\sqrt{x^2-1}}$.

c Comment on $f'(1)$ and $f'(-1)$.

d Use the expression for $f'(x)$ in part b to write down separate expressions for $f'(x)$ when $x > 1$ and when $x < -1$.

e Explain why $f(x)$ is increasing for $x > 1$ and for $x < -1$.

f Find (i) $\lim\limits_{x\to\infty} f(x)$ (ii) $\lim\limits_{x\to-\infty} f(x)$.

g Sketch the graph of $y = f(x)$.

ENRICHMENT

12 The function $f(x)$ is defined by the rule $f(x) = \sin^{-1}(\sin x)$.

a State the domain and range of $f(x)$, and whether it is even, odd or neither.

b Show that $f'(x) = \dfrac{\cos x}{|\cos x|}$.

c Is $f'(x)$ defined when $\cos x = 0$?

d What are the only two values that $f'(x)$ takes if $\cos x \neq 0$, and when does each of these values occur?

e Sketch the graph of $f(x)$ using the above information and a table of values if necessary.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **17 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 North Sydney Girls High School, 2023, HSC Trial Examination, Mathematics Extension 1, Q1 — 1 mark

Which of the following is the derivative of $y = \tan^{-1} 2x$?

A. $2\sec^2 2x$

B. $-2\tan^{-2} 2x$

C. $\dfrac{1}{1+4x^2}$

D. $\dfrac{2}{1+4x^2}$

Hint: use the linear extension with $a = 2$, $b = 0$.

PP 2 The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q4 — 1 mark

Which of the following is the derivative of $y = x^2 \sin^{-1}(2x)$?

(A) $2x\sin^{-1}(2x) + \dfrac{2x^2}{\sqrt{1-4x^2}}$

(B) $2x\sin^{-1}(2x) + \dfrac{x^2}{\sqrt{1-4x^2}}$

(C) $2x\sin^{-1}(2x) + \dfrac{2x^2}{\sqrt{1-2x^2}}$

(D) $2x\sin^{-1}(2x) + \dfrac{x^2}{\sqrt{1-2x^2}}$

Hint: product rule, then the linear extension for $\sin^{-1}(2x)$.

PP 3 Normanhurst Boys' High School, 2021, Mathematics Extension 1 (Year 12 Course) Assessment Task 4 (Trial Examination), Q5 — 1 mark

Which of the following is the derivative of $y = -\cos^{-1}(\ln x)$?

(A) $\dfrac{-1}{x\sqrt{1-2\ln x}}$

(B) $\dfrac{-1}{x\sqrt{(1+\ln x)(1-\ln x)}}$

(C) $\dfrac{1}{x\sqrt{1-2\ln x}}$

(D) $\dfrac{1}{x\sqrt{(1+\ln x)(1-\ln x)}}$

Hint: chain-rule extension with $f(x) = \ln x$, then factorise $1 - (\ln x)^2$.

PP 4 Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q4 — 1 mark

Which of the following is the equation of the tangent to $y = \cos^{-1} x$ at $x = 0$?

A. $2x + 2y + \pi = 0$

B. $2x + 2y - \pi = 0$

C. $2x - 2y + \pi = 0$

D. $2x - 2y - \pi = 0$

Hint: find the point and the gradient at $x = 0$, then rearrange $y - y_1 = m(x - x_1)$.

PP 5 Caringbah High School, 2020, Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(a) — 1 mark

Differentiate $\tan^{-1}(\log_e x)$.

Hint: $f(x) = \log_e x$ in the chain-rule extension.

PP 6 **Blacktown Boys' High School, 2022, HSC Trial Examination, Mathematics Extension 1, Q11(a)(i)** 2 marks

Differentiate $y = 3x \sin^{-1}(2x)$.

Hint: product rule with $u = 3x$ and $v = \sin^{-1}(2x)$.

PP 7 **Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q11(c)** 2 marks

Differentiate $y = x \cos^{-1} 4x$.

Hint: product rule; the derivative of $\cos^{-1} 4x$ is negative.

PP 8 **Gosford High School, 2022, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(c)** 2 marks

Differentiate $\log_e \left(\cos^{-1} x\right)$.

Hint: $\dfrac{d}{dx} \ln f(x) = \dfrac{f'(x)}{f(x)}$ with $f(x) = \cos^{-1} x$.

PP 9 **Killara High School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q12(d)** 3 marks

Find the equation of the tangent to the curve $y = \dfrac{1}{2} - \dfrac{1}{\pi} \sin^{-1}(2x)$ at the point $P\left(\dfrac{1}{4}, \dfrac{1}{3}\right)$.

Hint: differentiate, substitute $x = \frac{1}{4}$, and use $y - y_1 = m(x - x_1)$.

PP 10 **North Sydney Girls High School, 2023, HSC Trial Examination, Mathematics Extension 1, Q13(a)** 3 marks

Let $f(x) = \sin^{-1} x + \cos^{-1} x$.

(i) By differentiating $f(x)$, or otherwise, explain why $f(x) = \dfrac{\pi}{2}$. 2

(ii) Hence, sketch the graph of $y = f(x)$. 1

Hint: a function with zero derivative is constant; remember the domain when you sketch.

LESSON 12B

Inverse trigonometric functions: integrating

Reverse the standard derivatives to integrate $\frac{1}{\sqrt{a^2-x^2}}$ and $\frac{1}{a^2+x^2}$, handle the constants, use the reverse chain rule, and integrate the inverse trigonometric functions themselves.

CONCEPT 1 · the basic standard forms

REVERSING THE DERIVATIVES

Differentiation of the inverse trigonometric functions gave purely algebraic functions:

$$\frac{d}{dx}\sin^{-1}x=\frac{1}{\sqrt{1-x^2}} \quad \text{and} \quad \frac{d}{dx}\tan^{-1}x=\frac{1}{1+x^2}.$$

So some purely algebraic functions need the inverse trigonometric functions for their integration, just as $\int \frac{1}{x}\,dx=\log_e|x|$ needed the logarithm.

STANDARD FORMS FOR INTEGRATION

$$\int \frac{1}{\sqrt{1-x^2}}\,dx=\sin^{-1}x+C \quad \text{OR} \quad \int \frac{1}{\sqrt{1-x^2}}\,dx=-\cos^{-1}x+C \qquad \int \frac{1}{1+x^2}\,dx=\tan^{-1}x+C$$

Both forms of the first integral are correct: they differ by a constant, because $\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$.

TUTOR DEMO **Definite integrals**

Evaluate: **a** $\int_0^{\frac{\sqrt{3}}{2}} \frac{1}{\sqrt{1-x^2}}\,dx$ using both standard forms **b** $\int_0^{\sqrt{3}} \frac{1}{1+x^2}\,dx$

YOUR TURN **Same skills**

Evaluate: **a** $\int_0^{\frac{1}{\sqrt{2}}} \frac{1}{\sqrt{1-x^2}}\,dx$ **b** $\int_{-1}^{1} \frac{1}{1+x^2}\,dx$

WORKED EXAMPLE 1 **Keep for homework**

a Evaluate $\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}\,dx$ using both standard forms. **b** Evaluate $\int_0^4 \frac{1}{1+x^2}\,dx$ correct to four significant figures.

SOLUTION

$$\int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}} = \left[\sin^{-1} x\right]_0^{\frac{1}{2}}$$
$$= \sin^{-1}\tfrac{1}{2} - \sin^{-1} 0 = \tfrac{\pi}{6}$$

OR

$$\int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}} = \left[-\cos^{-1} x\right]_0^{\frac{1}{2}}$$
$$= -\tfrac{\pi}{3} + \tfrac{\pi}{2} = \tfrac{\pi}{6}$$

$$\int_0^4 \frac{dx}{1+x^2} = \left[\tan^{-1} x\right]_0^4$$
$$= \tan^{-1} 4 - \tan^{-1} 0$$
$$\approx 1.326$$

CONCEPT 2 · standard forms with constants

STANDARD FORMS WITH ONE CONSTANT

$$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \sin^{-1}\frac{x}{a} + C \quad \text{OR} \quad -\cos^{-1}\frac{x}{a} + C \qquad \int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$$

PROOF (FOR A > 0)

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \int \frac{dx}{a\sqrt{1-\left(\frac{x}{a}\right)^2}}$$
$$= \int \frac{1}{\sqrt{1-\left(\frac{x}{a}\right)^2}} \times \frac{1}{a}\,dx$$
$$= \sin^{-1}\frac{x}{a} + C$$

$$\int \frac{dx}{a^2+x^2} = \int \frac{dx}{a^2\left(1+\left(\frac{x}{a}\right)^2\right)}$$
$$= \frac{1}{a}\int \frac{1}{1+\left(\frac{x}{a}\right)^2} \times \frac{1}{a}\,dx$$
$$= \frac{1}{a}\tan^{-1}\frac{x}{a} + C$$

In each case let $u = \frac{x}{a}$, so $\frac{du}{dx} = \frac{1}{a}$, and use the basic standard form in u.

STANDARD FORMS WITH TWO CONSTANTS

$$\int \frac{1}{\sqrt{a^2-b^2x^2}}\,dx = \frac{1}{b}\sin^{-1}\frac{bx}{a} + C \quad \text{OR} \quad -\frac{1}{b}\cos^{-1}\frac{bx}{a} + C \qquad \int \frac{1}{a^2+b^2x^2}\,dx = \frac{1}{ab}\tan^{-1}\frac{bx}{a} + C$$

If you prefer the one-constant forms, take the coefficient of x^2 out first, as in parts c and d of Worked Example 2.

TUTOR DEMO **One constant, then two**

Find: **a** $\int \frac{3}{\sqrt{16-x^2}}\,dx$ **b** $\int \frac{1}{25+4x^2}\,dx$

YOUR TURN **Same skills**

Find: a $\int \frac{2}{\sqrt{25-x^2}}\,dx$ b $\int \frac{1}{9+4x^2}\,dx$

WORKED EXAMPLE 2 **Keep for homework**

Find: a $\int \frac{2}{\sqrt{9-x^2}}\,dx$ b $\int \frac{1}{8+x^2}\,dx$ c $\int \frac{6}{49+25x^2}\,dx$ d $\int \frac{1}{\sqrt{5-3x^2}}\,dx$

SOLUTION

In parts a and b the formulae apply immediately; in c and d take out the coefficient of x^2 first.

a $\int \frac{2}{\sqrt{9-x^2}}\,dx = 2\sin^{-1}\frac{x}{3} + C$

b $\int \frac{1}{8+x^2}\,dx = \frac{1}{2\sqrt{2}}\tan^{-1}\frac{x}{2\sqrt{2}} + C$

c
$$\int \frac{6}{49+25x^2}\,dx = \frac{6}{25}\int \frac{1}{\frac{49}{25}+x^2}\,dx$$
$$= \frac{6}{25}\times\frac{5}{7}\tan^{-1}\frac{x}{7/5} + C$$
$$= \frac{6}{35}\tan^{-1}\frac{5x}{7} + C$$

d
$$\int \frac{dx}{\sqrt{5-3x^2}} = \frac{1}{\sqrt{3}}\int \frac{dx}{\sqrt{\frac{5}{3}-x^2}}$$
$$= \tfrac{1}{3}\sqrt{3}\,\sin^{-1}\left(x\sqrt{\tfrac{3}{5}}\right) + C$$

CONCEPT 3 · the reverse chain rule

REVERSE CHAIN RULE FORMS

$$\int \frac{f'(x)}{\sqrt{1-\left(f(x)\right)^2}}\,dx = \sin^{-1} f(x) + C \quad \text{OR} \quad -\cos^{-1} f(x) + C$$

$$\int \frac{f'(x)}{1+\left(f(x)\right)^2}\,dx = \tan^{-1} f(x) + C$$

These are the reversals of the chain-rule forms at the end of Lesson 12A. Look for a function whose derivative also appears in the numerator; adjust constants as needed.

TUTOR DEMO **Spot f(x) and f'(x)**

Find primitives of: a $\frac{e^{2x}}{1+e^{4x}}$ b $\frac{\cos x}{\sqrt{1-\sin^2 x}}$, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

YOUR TURN **Same method**

Find a primitive of $\dfrac{x}{1+x^4}$.

WORKED EXAMPLE 3 **Keep for homework**

Find primitives of: **a** $\dfrac{e^x}{1+e^{2x}}$ **b** $\dfrac{x}{\sqrt{1-x^4}}$

SOLUTION

a Let $f(x)=e^x$, then $f'(x)=e^x$:

$$\int \frac{e^x}{1+(e^x)^2}\,dx = \tan^{-1} e^x + C$$

b Let $f(x)=x^2$, then $f'(x)=2x$:

$$\int \frac{x}{\sqrt{1-x^4}}\,dx = \tfrac{1}{2}\int \frac{2x}{\sqrt{1-(x^2)^2}}\,dx = \tfrac{1}{2}\sin^{-1} x^2 + C$$

CONCEPT 4 · given a derivative, find an integral

USING A PRODUCT-RULE DERIVATIVE

As always, the result of a differentiation can be used to obtain an integral. In particular, differentiating $x\sin^{-1}x$ (or $x\tan^{-1}x$) with the product rule allows the primitives of the inverse trigonometric functions themselves to be found.

TUTOR DEMO **Primitive of $\tan^{-1}x$**

a Differentiate $x\tan^{-1}x$. **b** Hence find $\displaystyle\int_0^1 \tan^{-1}x\,dx$.

YOUR TURN **Primitive of $\cos^{-1}x$**

a Show that $\dfrac{d}{dx}\left(x\cos^{-1}x-\sqrt{1-x^2}\right)=\cos^{-1}x$. **b** Hence find $\displaystyle\int_0^1 \cos^{-1}x\,dx$.

WORKED EXAMPLE 4 **Keep for homework**

a Differentiate $x\sin^{-1}x$, and hence find a primitive of $\sin^{-1}x$. **b** Find the shaded area under the curve $y=\sin^{-1}x$ from $x=0$ to $x=1$.

SOLUTION

a Using the product rule with $u=x$ and $v=\sin^{-1}x$:

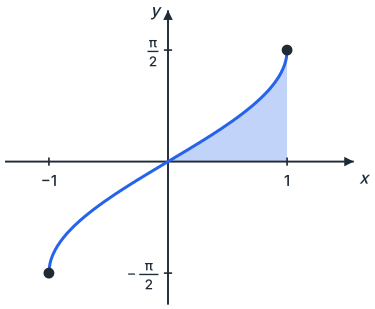
$$\frac{d}{dx}\left(x\sin^{-1}x\right)=\sin^{-1}x+\frac{x}{\sqrt{1-x^2}}.$$

Hence $\displaystyle\int\sin^{-1}x\,dx=x\sin^{-1}x-\int\frac{x}{\sqrt{1-x^2}}\,dx.$

By the reverse chain rule ($u=1-x^2$, $\frac{du}{dx}=-2x$):

$$-\int\frac{x}{\sqrt{1-x^2}}\,dx=\tfrac{1}{2}\int(1-x^2)^{-\frac{1}{2}}(-2x)\,dx=\sqrt{1-x^2},$$

so $\displaystyle\int\sin^{-1}x\,dx=x\sin^{-1}x+\sqrt{1-x^2}+C.$



b $\displaystyle\int_0^1\sin^{-1}x\,dx=\left[x\sin^{-1}x+\sqrt{1-x^2}\right]_0^1=\left(\tfrac{\pi}{2}+0\right)-(0+1)=\tfrac{\pi}{2}-1$ **square units.**

Check: the area between $y=\sin^{-1}x$ and the y-axis is the same as the area under $y=\sin x$ from 0 to $\frac{\pi}{2}$, which is 1. Subtracting from the rectangle of area $\frac{\pi}{2}$ gives $\frac{\pi}{2}-1$ again.

TEXTBOOK HOMEWORK · Exercise 12B

FOUNDATION

1 Find:

a $\int \frac{-1}{\sqrt{1-x^2}}\,dx$ b $\int \frac{1}{\sqrt{4-x^2}}\,dx$ c $\int \frac{1}{9+x^2}\,dx$

d $\int \frac{1}{\sqrt{\frac{4}{9}-x^2}}\,dx$ e $\int \frac{1}{2+x^2}\,dx$ f $\int \frac{-1}{\sqrt{5-x^2}}\,dx$

2 Find the exact value of:

a $\int_0^3 \frac{1}{\sqrt{9-x^2}}\,dx$ b $\int_0^2 \frac{1}{4+x^2}\,dx$ c $\int_0^1 \frac{1}{\sqrt{2-x^2}}\,dx$

d $\int_{\frac{1}{2}}^{\frac{1}{2}\sqrt{3}} \frac{\frac{1}{2}}{\frac{1}{4}+x^2}\,dx$ e $\int_{\frac{1}{6}\sqrt{3}}^{\frac{1}{6}} \frac{-1}{\sqrt{\frac{1}{9}-x^2}}\,dx$ f $\int_{-\frac{3}{4}\sqrt{2}}^{\frac{3}{4}} \frac{1}{\sqrt{\frac{9}{4}-x^2}}\,dx$

3 Find the equation of the curve, given that:

a $y' = (1-x^2)^{-\frac{1}{2}}$ and the curve passes through the point $(0, \pi)$.

b $y' = 4(16+x^2)^{-1}$ and the curve passes through the point $(-4, 0)$.

4

a If $y' = \frac{1}{\sqrt{36-x^2}}$ and $y = \frac{\pi}{6}$ when $x = 3$, find the value of y when $x = 3\sqrt{3}$.

b Given that $y' = \frac{2}{4+x^2}$ and that $y = \frac{\pi}{3}$ when $x = 2$, find y when $x = \frac{2}{\sqrt{3}}$.

DEVELOPMENT

5 Find:

a $\displaystyle\int \frac{1}{\sqrt{1-4x^2}}\,dx$

b $\displaystyle\int \frac{1}{1+16x^2}\,dx$

c $\displaystyle\int \frac{-1}{\sqrt{1-2x^2}}\,dx$

d $\displaystyle\int \frac{1}{\sqrt{4-9x^2}}\,dx$

e $\displaystyle\int \frac{1}{25+9x^2}\,dx$

f $\displaystyle\int \frac{-1}{\sqrt{3-4x^2}}\,dx$

6 Find the exact value of:

a $\displaystyle\int_0^{\frac{1}{6}} \frac{1}{\sqrt{1-9x^2}}\,dx$

b $\displaystyle\int_{\frac{1}{2}}^{\frac{1}{2}\sqrt{3}} \frac{2}{1+4x^2}\,dx$

c $\displaystyle\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\sqrt{1-3x^2}}\,dx$

d $\displaystyle\int_{-\frac{3}{4}}^{\frac{3}{2\sqrt{2}}} \frac{1}{\sqrt{9-4x^2}}\,dx$

7 a Shade the region bounded by $y = \sin^{-1} x$, the x-axis and the vertical line $x = \frac{1}{2}$.

b Show that $\dfrac{d}{dx}\left(x\sin^{-1}x + \sqrt{1-x^2}\right) = \sin^{-1}x$.

c Hence find the exact area of the region.

8 a Show that $\dfrac{d}{dx}\left(\cos^{-1}(2-x)\right) = \dfrac{1}{\sqrt{4x-x^2-3}}$.

b Hence find $\displaystyle\int_1^2 \frac{1}{\sqrt{4x-x^2-3}}\,dx$.

9 a Differentiate $\tan^{-1}\frac{1}{2}x^3$.

b Hence find $\displaystyle\int \frac{x^2}{4+x^6}\,dx$.

10 Without finding any primitives, use arguments from symmetry or geometry to evaluate:

a $\displaystyle\int_{-\frac{1}{3}}^{\frac{1}{3}} \sin^{-1}x\,dx$ b $\displaystyle\int_{-5}^{5} \tan^{-1}x\,dx$ c $\displaystyle\int_{-\frac{3}{4}}^{\frac{3}{4}} \cos^{-1}x\,dx$

d $\displaystyle\int_{-\frac{2}{3}}^{\frac{2}{3}} \frac{x}{\sqrt{1-x^2}}\,dx$ e $\displaystyle\int_{-3}^{3} \frac{x}{1+x^2}\,dx$ f $\displaystyle\int_{-6}^{6} \sqrt{36-x^2}\,dx$

11 a Given that $f(x)=\dfrac{x}{1+x^2}-\tan^{-1}x$: (i) find $f(0)$; (ii) show that $f'(x)=\dfrac{-2x^2}{(1+x^2)^2}$.

b Hence: (i) explain why $f(x)<0$ for all $x>0$; (ii) find $\displaystyle\int_0^1 \frac{x^2}{(1+x^2)^2}\,dx$.

12 Consider the function $f(x)=\dfrac{1}{\sqrt{4-x^2}}$.

a Sketch the graph of $y=\sqrt{4-x^2}$.

b Hence sketch the graph of $y=f(x)$.

c Find the area between the curve and the x-axis from $x=-1$ to $x=1$.

d Find the total area between the curve and the x-axis. [This is an example of an unbounded region having a finite area.]

13 Consider the function $f(x) = \dfrac{4}{x^2+4}$.

a What is the axis of symmetry of $y = f(x)$?

b What are the domain and range?

c Show that the graph of $f(x)$ has a maximum turning point at $(0, 1)$.

d Find $\lim\limits_{x\to\infty} f(x)$, and hence sketch $y = f(x)$. On the same axes, sketch $y = \frac{1}{4}(x^2+4)$.

e Calculate the area bounded by the curve and the x-axis from $x = -2\sqrt{3}$ to $x = \frac{2}{3}\sqrt{3}$.

f Find the exact area between the curve and the x-axis from $x = -a$ to $x = a$, where a is a positive constant.

g By letting a tend to infinity, find the total area between the curve and the x-axis.

ENRICHMENT

14 a Show that $\dfrac{d}{dx}\left(\tan^{-1}\left(\frac{3}{2}\tan x\right)\right) = \dfrac{6}{5\sin^2 x + 4}$.

b Hence find the area under the curve $y = \dfrac{1}{5\sin^2 x + 4}$ from $x = 0$ to $x = 7$.

c Why is the calculation asked for in part b invalid and completely wrong?

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **18 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 Normanhurst Boys' High School, 2022, Mathematics Extension 1 Assessment Task 4 (Trial Examination), Q8 — 1 mark

What is the primitive of $\dfrac{2f'(x)}{1+4\left(f(x)\right)^2}$?

(A) $\tan^{-1}\left(f(2x)\right)$
(B) $2\tan^{-1}\left(f(x)\right)$
(C) $2\ln 2\left(f(x)\right)$
(D) $\tan^{-1}\left(2f(x)\right)$

Hint: write $4\left(f(x)\right)^2$ as $\left(2f(x)\right)^2$. What is the derivative of $2f(x)$?

PP 2 Sydney Grammar School, 2021, Trial Examination, Form VI Mathematics Extension 1, Q12(d) — 2 marks

Find $\displaystyle\int \frac{1}{9+16x^2}\,dx$.

Hint: use the two-constant form with $a=3$, $b=4$, or take out 16 first.

PP 3 Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q11(a) — 2 marks

Find $\displaystyle\int \frac{1}{\sqrt{4-9x^2}}\,dx$.

Hint: two-constant form with $a=2$, $b=3$.

PP 4 Blacktown Boys' High School, 2022, HSC Trial Examination, Mathematics Extension 1, Q11(a)(ii) — 2 marks

Evaluate $\displaystyle\int_0^{\frac{1}{3}} \frac{-1}{\sqrt{\frac{1}{9}-x^2}}\,dx$.

Hint: the negative numerator suggests the $\cos^{-1}$ form, with $a=\frac{1}{3}$.

PP 5 Gosford High School, 2022, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(f) — 2 marks

Find the equation of the curve $f(x)$ that passes through the point $\left(0,-\dfrac{\pi}{2}\right)$ and has $f'(x)=-\dfrac{3}{\sqrt{16-9x^2}}$.

Hint: integrate, then substitute the point to find the constant.

PP 6 **Fort Street High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q12(d)** 4 marks

(i) Differentiate $x\sin^{-1}x+\sqrt{1-x^2}$. 2

(ii) Hence evaluate $\displaystyle\int_0^1 2\sin^{-1}x\,dx$. 2

Hint: this is Worked Example 4 again; use the result of (i) as the primitive.

PP 7 **Hornsby Girls High School, 2022, Year 12 Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)** 5 marks

(i) Show that $\dfrac{d}{dx}\left(\dfrac{2x}{4+x^2}+\tan^{-1}\dfrac{x}{2}\right)=\dfrac{16}{(4+x^2)^2}$. 3

(ii) Hence, evaluate $\displaystyle\int_0^{2\sqrt{3}}\frac{dx}{(4+x^2)^2}$. 2

Hint: quotient rule for the first term; put everything over $(4+x^2)^2$. In (ii), divide the result of (i) by 16.

LESSON 12C

Further trigonometric integrals

Integrate $\sin^2 x$ and $\cos^2 x$ using the $\cos 2\theta$ formulae, integrate the squares of the other trigonometric functions, and use the reverse chain rule and products-to-sums with trigonometric integrands.

CONCEPT 1 · six standard integrals

SIX STANDARD INTEGRALS

$$\int \cos x\,dx = \sin x + C \qquad \int \sin x\,dx = -\cos x + C \qquad \int \sec^2 x\,dx = \tan x + C$$

$$*\int \operatorname{cosec}^2 x\,dx = -\cot x + C \qquad *\int \sec x \tan x\,dx = \sec x + C \qquad *\int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + C$$

The three marked * need not be memorised: they are reversals of derivatives you can check by differentiating.

CONCEPT 2 · the primitives of sin²x and cos²x

USE THE COS 2Θ FORMULAE FROM LESSON 11A

$$\sin^2\theta = \tfrac{1}{2} - \tfrac{1}{2}\cos 2\theta \qquad \cos^2\theta = \tfrac{1}{2} + \tfrac{1}{2}\cos 2\theta$$

It is better to remember these identities than the actual integrals. Replace the square, then integrate term by term.

INTEGRATING THE SQUARES OF THE TRIGONOMETRIC FUNCTIONS

$$\int \sin^2 x\,dx = \int \left(\tfrac{1}{2} - \tfrac{1}{2}\cos 2x\right)dx \qquad \int \cos^2 x\,dx = \int \left(\tfrac{1}{2} + \tfrac{1}{2}\cos 2x\right)dx$$

$$\int \sec^2 x\,dx = \tan x + C \qquad \int \tan^2 x\,dx = \int \left(\sec^2 x - 1\right)dx$$

$$*\int \operatorname{cosec}^2 x\,dx = -\cot x + C \qquad *\int \cot^2 x\,dx = \int \left(\operatorname{cosec}^2 x - 1\right)dx$$

TUTOR DEMO **Replace the square**

Evaluate: **a** $\displaystyle\int_0^{\frac{\pi}{4}} \cos^2 x\,dx$ **b** $\displaystyle\int_0^{\pi} \sin^2 \tfrac{1}{2}x\,dx$

YOUR TURN **Same method**

Evaluate: **a** $\displaystyle\int_0^{\frac{\pi}{2}} \cos^2 x\,dx$ **b** $\displaystyle\int_0^{\frac{\pi}{2}} \sin^2 2x\,dx$

TUTOR DEMO **tan²x**

Find $\int \tan^2 x \, dx$.

YOUR TURN **A definite version**

Evaluate $\int_0^{\frac{\pi}{4}} \tan^2 x \, dx$.

WORKED EXAMPLE 1 **Keep for homework**

Evaluate: a $\int_0^{\frac{\pi}{2}} \sin^2 x \, dx$ b $\int_0^{\frac{2\pi}{3}} \cos^2 \frac{1}{2}x \, dx$

SOLUTION

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^2 x \, dx &= \int_0^{\frac{\pi}{2}} \left(\tfrac{1}{2} - \tfrac{1}{2}\cos 2x\right) dx \\ &= \left[\tfrac{1}{2}x - \tfrac{1}{4}\sin 2x\right]_0^{\frac{\pi}{2}} \\ &= \left(\tfrac{\pi}{4} - 0\right) - (0 - 0) \\ &= \tfrac{\pi}{4}\end{aligned}$$

$$\begin{aligned}\int_0^{\frac{2\pi}{3}} \cos^2 \tfrac{1}{2}x \, dx &= \int_0^{\frac{2\pi}{3}} \left(\tfrac{1}{2} + \tfrac{1}{2}\cos x\right) dx \\ &= \left[\tfrac{1}{2}x + \tfrac{1}{2}\sin x\right]_0^{\frac{2\pi}{3}} \\ &= \left(\tfrac{\pi}{3} + \tfrac{\sqrt{3}}{4}\right) - (0 + 0) \\ &= \tfrac{\pi}{3} + \tfrac{\sqrt{3}}{4}\end{aligned}$$

In part b the angle is $\frac{1}{2}x$, so $\cos 2\theta$ becomes $\cos x$.

CONCEPT 3 · trigonometric integrals using the reverse chain rule

REVERSE CHAIN RULE WITH TRIGONOMETRIC FUNCTIONS

$$\int \frac{f'(x)}{f(x)} \, dx = \log_e |f(x)| + C \qquad \int f'(x)\big(f(x)\big)^n \, dx = \frac{\big(f(x)\big)^{n+1}}{n+1} + C$$

Look for $\sin x$ with $\cos x$ (one is the derivative of the other, up to sign), or $\tan x$ with $\sec^2 x$. This prepares for substitution in Lessons 12D and 12E.

TUTOR DEMO **Log and power forms**

Find: a $\int \tan x \, dx$ b $\int \cos x \sin^3 x \, dx$

YOUR TURN **Same method**

Find: **a** $\int \cot 3x\,dx$ **b** $\int \sin x \cos^5 x\,dx$

WORKED EXAMPLE 2 **Keep for homework**

a Integrate $\cot x$ using the reverse chain rule. **b** Find primitives of $\sin x \cos^4 x$ and $\cos x \sin^n x$.

SOLUTION

a $\displaystyle\int \cot x\,dx = \int \frac{\cos x}{\sin x}\,dx = \log_e |\sin x| + C \qquad (u = \sin x,\ u' = \cos x)$

b $\displaystyle\int \sin x \cos^4 x\,dx = -\int (-\sin x)\cos^4 x\,dx = -\tfrac{1}{5}\cos^5 x + C \qquad (u = \cos x,\ u' = -\sin x)$

$\displaystyle\int \cos x \sin^n x\,dx = \frac{\sin^{n+1} x}{n+1} + C \qquad (u = \sin x,\ u' = \cos x,\ n \neq -1)$

CONCEPT 4 · products to sums

INTEGRATING A PRODUCT OF TWO TRIG FUNCTIONS

Use the products-to-sums formulae from the Chapter 11 formula sheet to turn a product such as $\sin 5x \cos 3x$ into a sum, then integrate each term:

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$
$$2\cos A\cos B = \cos(A+B) + \cos(A-B)$$
$$-2\sin A\sin B = \cos(A+B) - \cos(A-B)$$

TUTOR DEMO **sin × cos**

Find $\displaystyle\int \sin 5x \cos 3x\,dx$.

YOUR TURN **cos × cos**

Find $\displaystyle\int 2\cos 3x \cos x\,dx$.

TEXTBOOK HOMEWORK · Exercise 12C

FOUNDATION

1 Use double-angle results to show that:

a $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$

b $\cos^2 2x = \frac{1}{2} + \frac{1}{2}\cos 4x$

c $\sin 3x \cos 3x = \frac{1}{2}\sin 6x$

d $2\sin^2 \frac{x}{2} = 1 - \cos x$

2 Without a calculator, find the value of:

a $\cos^2 15°$

b $\sin^2 \frac{5\pi}{12}$

c $\sin 105° \cos 105°$

d $\sin^2 \frac{7\pi}{8}$

3 Express $\sin^2 \theta$ in terms of $\cos 2\theta$, and hence find:

a $\int \sin^2 x \, dx$

b $\int \sin^2 2x \, dx$

c $\int \sin^2 \frac{1}{4}x \, dx$

d $\int \sin^2 3x \, dx$

4 Express $\cos^2 \theta$ in terms of $\cos 2\theta$, and hence find:

a $\int \cos^2 x \, dx$

b $\int \cos^2 6x \, dx$

c $\int \cos^2 \frac{1}{2}x \, dx$

d $\int \cos^2 10x \, dx$

DEVELOPMENT

5 Use the results $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$ and $\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$ to evaluate:

a $\int_0^{\pi} \sin^2 x \, dx$

b $\int_0^{\frac{\pi}{4}} \cos^2 x \, dx$

c $\int_0^{\frac{\pi}{6}} \sin^2 \frac{1}{2}x \, dx$

d $\int_0^{\frac{\pi}{16}} \cos^2 2x \, dx$

e $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos^2\left(x + \frac{\pi}{12}\right) dx$

f $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin^2\left(x - \frac{\pi}{6}\right) dx$

6 Use the products-to-sums formulae to find:

a $\int \sin 3x \cos 2x \, dx$

b $\int \cos 3x \sin x \, dx$

c $\int_0^{\frac{\pi}{4}} 2\cos 2x \cos x \, dx$

d $\int_0^{\frac{\pi}{3}} \sin 5x \sin 2x \, dx$

7 a Sketch the graph of $y = \cos 2x$, for $0 \leq x \leq 2\pi$.

b Hence sketch, on the same diagram, $y = \frac{1}{2}(1 + \cos 2x)$ and $y = \frac{1}{2}(1 - \cos 2x)$.

c Deduce graphically that $\cos^2 x + \sin^2 x = 1$.

8 Use the reverse chain rule to find:

a $\int \sin^3 x \cos x \, dx$

b $\int \sin^6 x \cos x \, dx$

c $\int \cos^5 x \sin x \, dx$

d $\int \cos^8 x \sin x \, dx$

e $\int e^x \sin e^x \, dx$

f $\int e^x \cos 5e^x \, dx$

g $\int \tan x \, dx$ (let $u = \cos x$)

h $\int \cot 7x \, dx$

9 a Find the range of the function $y = \cos x \sin x$.

b Integrate $y = \cos x \sin x$ in three ways: (i) using the $\sin 2x$ formula, (ii) using the reverse chain rule in two different ways.

c Reconcile the three results.

10 a By writing $\sin^4 x$ as $(\sin^2 x)^2$, show that $\sin^4 x = \frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$.

b Find a similar result for $\cos^4 x$.

c Hence find: (i) $\int_0^{\pi} \sin^4 x \, dx$ (ii) $\int_0^{\frac{\pi}{4}} \cos^4 x \, dx$

d Use the Pythagorean identity to find $\int_0^{\frac{\pi}{3}} \sin^3 x \, dx$.

11 Evaluate using the standard forms $\int \sec^2 x\,dx = \tan x$ and $\int \operatorname{cosec}^2 x\,dx = -\cot x$:

a $\displaystyle\int \tan^2 2x\,dx$

b $\displaystyle\int \cot^2 \tfrac{1}{2}x\,dx$

c $\displaystyle\int_{\frac{\pi}{12}}^{\frac{\pi}{9}} 3\tan^2 3x\,dx$

d $\displaystyle\int_{\frac{\pi}{24}}^{\frac{\pi}{8}} \cot^2 4x\,dx$

12 Integrate these functions, using the reverse chain rule or otherwise:

a $\tan x \sec^2 x$

b $\dfrac{\sin^2 x}{1+\cos x}$

c $\dfrac{1+\cos^3 x}{\cos^2 x}$

ENRICHMENT

13 Define $F(x) = \displaystyle\int_0^x \sin^2 t\,dt$, where $0 \le x \le 2\pi$.

a Show that $F(x) = \frac{1}{2}x - \frac{1}{4}\sin 2x$.

b Explain why $F'(x) = \sin^2 x$. Hence state the values of x in the given domain for which $F(x)$ is: (i) stationary, (ii) increasing, (iii) decreasing.

c Explain why $F(x)$ never differs from $\frac{1}{2}x$ by more than $\frac{1}{4}$.

d Find any points of inflection of $F(x)$ in the given domain.

e Sketch, on the same diagram, the graphs of $y = F(x)$ and $y = F'(x)$ over the given domain, and observe how they are related.

f (i) For what value of k is $\displaystyle\int_0^k \sin^2 x\,dx = \frac{3\pi}{2}$? (ii) For what values of k is $\displaystyle\int_0^k \sin^2 x\,dx = \frac{n\pi}{2}$, where n is an integer?

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **18 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q1 — 1 mark

Which of the following is an expression for $\int \sin^2 2x \, dx$?

A. $\frac{x}{2} - \frac{1}{8}\sin 4x + C$ B. $\frac{x}{2} + \frac{1}{8}\sin 4x + C$ C. $\frac{x}{2} - \frac{1}{4}\sin 4x + C$ D. $\frac{x}{2} + \frac{1}{4}\sin 4x + C$

Hint: $\sin^2 2x = \frac{1}{2} - \frac{1}{2}\cos 4x$.

PP 2 North Sydney Girls High School, 2023, HSC Trial Examination, Mathematics Extension 1, Q3 — 1 mark

Which of the following is equal to $2\sin 3x \cos 4x$?

A. $\sin 7x - \sin x$ B. $\sin 7x + \sin x$ C. $\cos 7x - \cos x$ D. $\cos x - \cos 7x$

Hint: $2\sin A \cos B = \sin(A+B) + \sin(A-B)$, and $\sin(-x) = -\sin x$.

PP 3 Normanhurst Boys' High School, 2022, Mathematics Extension 1 Assessment Task 4 (Trial Examination), Q13(b) — 2 marks

Find $\int \sin^2 3x \, dx$.

Hint: $\sin^2 3x = \frac{1}{2} - \frac{1}{2}\cos 6x$.

PP 4 Sydney Grammar School, 2021, Trial Examination, Form VI Mathematics Extension 1, Q14(a) — 2 marks

Use double angle formulae to find $\int \sin^2 x \cos^2 x \, dx$.

Hint: $\sin x \cos x = \frac{1}{2}\sin 2x$, so the integrand is $\frac{1}{4}\sin^2 2x$.

PP 5 Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q12(e) — 3 marks

Write $\cos^2 2x$ in terms of $\cos 4x$ and hence evaluate $\int_0^{\frac{\pi}{6}} \cos^2 2x \, dx$.

Hint: $\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$ with $\theta = 2x$.

PP 6 **Cherrybrook Technology High School, 2020, Year 12 AP4 Mathematics Extension 1 (Trial), Q13(a)** 2 marks

Find $\int_0^{2\pi} \sin^2\left(\frac{x}{2}\right) dx$.

Hint: the double angle of $\frac{x}{2}$ is x.

PP 7 **Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q11(d)** 3 marks

Find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos^2 x \, dx$.

Hint: after integrating, notice that $\sin\frac{2\pi}{3} = \sin\frac{\pi}{3}$.

PP 8 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(a)** 2 marks

Find $\int_0^{\frac{\pi}{2}} \sin^2 x \cos x \, dx$.

Hint: this is not a $\cos 2x$ question: $\cos x$ is the derivative of $\sin x$.

PP 9 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(a)** 2 marks

Find $\int \sin 8x \cos 5x \, dx$.

Hint: products to sums with $A = 8x$, $B = 5x$.

LESSON 12D

Integration by substitution

Rewrite the reverse chain rule in the notation of substitution, "let $u =$ some function of x", and change the limits of a definite integral so there is no need to return to x.

CONCEPT 1 · from the reverse chain rule to substitution

THE NEW NOTATION

The reverse chain rule does not cover every situation where the chain rule can be used in integration. Integration by substitution is a more general method. Its first stage uses substitutions of the form $\textbf{Let } u = \textbf{some function of } x$.

The key to the notation is that the derivative $\dfrac{du}{dx}$ is treated as a fraction: the du and the dx are split apart, so that $\dfrac{du}{dx} = -2x$ is written instead as $du = -2x\,dx$. The new variable u then appears in the main line of working, not only in a side column.

THE METHOD

1. Write down u and find du in terms of dx.
2. Replace every x expression in the integrand (and the dx) by u expressions (and du).
3. Integrate with respect to u.
4. For an indefinite integral, substitute back to x.

WORKED EXAMPLE 1 The same integral, two notations

Find $\displaystyle\int x(1-x^2)^4\,dx$: **a** by the reverse chain rule, **b** using the substitution $u = 1 - x^2$.

SOLUTION

a Reverse chain rule ($u = 1 - x^2$, $\frac{du}{dx} = -2x$ in the side working):

$$\begin{aligned}\int x(1-x^2)^4\,dx &= -\tfrac{1}{2}\int(-2x)(1-x^2)^4\,dx \\ &= -\tfrac{1}{2}\times\tfrac{1}{5}(1-x^2)^5 + C \\ &= -\tfrac{1}{10}(1-x^2)^5 + C\end{aligned}$$

b Let $u = 1 - x^2$. Then $du = -2x\,dx$, and $x\,dx = -\frac{1}{2}\,du$.

$$\begin{aligned}\int x(1-x^2)^4\,dx &= \int u^4\left(-\tfrac{1}{2}\right)du \\ &= -\tfrac{1}{2}\times\tfrac{1}{5}u^5 + C \\ &= -\tfrac{1}{10}(1-x^2)^5 + C\end{aligned}$$

TUTOR DEMO A given substitution

Find $\displaystyle\int x^2(1+x^3)^4\,dx$, using the substitution $u = 1 + x^3$.

YOUR TURN Same method

Find $\displaystyle\int x(x^2+3)^5\,dx$, using the substitution $u = x^2 + 3$.

CONCEPT 2 · an advance on the reverse chain rule

WHEN THE REVERSE CHAIN RULE IS CLUMSY

Some integrals that can be done this way could only be done by the reverse chain rule in a rather clumsy manner. If, after replacing dx, some x is left over, use the equation for u to write that x in terms of u as well (for example $u = 1 - x$ gives $x = 1 - u$).

TUTOR DEMO **Trigonometric substitution u = f(x)**

Find $\int \cos x\sqrt{1+\sin x}\,dx$, using the substitution $u = 1 + \sin x$.

TUTOR DEMO **An x left over**

Find $\int x\sqrt{x+2}\,dx$, using the substitution $u = x + 2$.

YOUR TURN **Same method**

Find $\int x\sqrt{x-1}\,dx$, using the substitution $u = x - 1$.

WORKED EXAMPLE 2 **Keep for homework**

Find: **a** $\int \sin x\sqrt{1-\cos x}\,dx$, using $u = 1 - \cos x$ **b** $\int x\sqrt{1-x}\,dx$, using $u = 1 - x$

SOLUTION

a Let $u = 1 - \cos x$. Then $du = \sin x\,dx$.

$$\begin{aligned}\int \sin x\sqrt{1-\cos x}\,dx &= \int u^{\frac{1}{2}}\,du \\ &= \tfrac{2}{3}u^{\frac{3}{2}} + C \\ &= \tfrac{2}{3}(1-\cos x)^{\frac{3}{2}} + C\end{aligned}$$

b Let $u = 1 - x$. Then $du = -dx$, and $x = 1 - u$.

$$\begin{aligned}\int x\sqrt{1-x}\,dx &= \int (1-u)\sqrt{u}\,(-du) \\ &= \int \left(u^{\frac{3}{2}} - u^{\frac{1}{2}}\right)du \\ &= \tfrac{2}{5}u^{\frac{5}{2}} - \tfrac{2}{3}u^{\frac{3}{2}} + C \\ &= \tfrac{2}{5}(1-x)^{\frac{5}{2}} - \tfrac{2}{3}(1-x)^{\frac{3}{2}} + C\end{aligned}$$

CONCEPT 3 · substituting the limits of integration

DEFINITE INTEGRALS

A great advantage of the new method is that the limits of integration can be changed from values of x to values of u. There is then no need ever to go back to x.

Write the new limits in the side working: "When $x = \ldots$, $u = \ldots$". The lower limit stays at the bottom of the integral even if it is the larger number.

TUTOR DEMO New limits

Evaluate $\displaystyle\int_0^1 \frac{x^2}{(1+x^3)^2}\,dx$, using the substitution $u = 1 + x^3$.

YOUR TURN Same method

Evaluate $\displaystyle\int_0^{\frac{\pi}{3}} \sin x \cos^3 x\,dx$, using the substitution $u = \cos x$.

WORKED EXAMPLE 3 Keep for homework

Find: **a** $\displaystyle\int_0^1 x\sqrt{1-x}\,dx$, using $u = 1 - x$ **b** $\displaystyle\int_0^{\pi} \sin x \cos^6 x\,dx$, using $u = \cos x$

SOLUTION

a Let $u = 1 - x$. Then $du = -dx$ and $x = 1 - u$. When $x = 0$, $u = 1$; when $x = 1$, $u = 0$.

$$\begin{aligned}\int_0^1 x\sqrt{1-x}\,dx &= -\int_1^0 (1-u)\sqrt{u}\,du \\ &= -\int_1^0 \left(u^{\frac{1}{2}} - u^{\frac{3}{2}}\right)du \\ &= -\left[\tfrac{2}{3}u^{\frac{3}{2}} - \tfrac{2}{5}u^{\frac{5}{2}}\right]_1^0 \\ &= -0 + \left(\tfrac{2}{3} - \tfrac{2}{5}\right) = \tfrac{4}{15}\end{aligned}$$

b Let $u = \cos x$. Then $du = -\sin x\,dx$. When $x = 0$, $u = 1$; when $x = \pi$, $u = -1$.

$$\begin{aligned}\int_0^{\pi} \sin x \cos^6 x\,dx &= -\int_1^{-1} u^6\,du \\ &= -\tfrac{1}{7}\left[u^7\right]_1^{-1} \\ &= -\tfrac{1}{7} \times (-1) + \tfrac{1}{7} \times 1 \\ &= \tfrac{2}{7}\end{aligned}$$

TEXTBOOK HOMEWORK · Exercise 12D

FOUNDATION

1 Consider the integral $\int 2x(1+x^2)^3\,dx$, and the substitution $u = 1 + x^2$.

a Show that $du = 2x\,dx$.

b Show that the integral can be written as $\int u^3\,du$.

c Hence find the primitive of $2x(1+x^2)^3$.

d Check your answer by differentiating it.

2 Repeat the previous question for each indefinite integral and substitution.

a $\int 2(2x+3)^3\,dx$ (Let $u = 2x + 3$.)

b $\int \frac{2x}{(1+x^2)^2}\,dx$ (Let $u = 1 + x^2$.)

c $\int \frac{3}{\sqrt{3x-5}}\,dx$ (Let $u = 3x - 5$.)

d $\int \frac{4x^3}{1+x^4}\,dx$ (Let $u = 1 + x^4$.)

3 Consider the integral $\int \frac{x}{\sqrt{1-x^2}}\,dx$, and the substitution $u = 1 - x^2$.

a Show that $x\,dx = -\frac{1}{2}\,du$.

b Show that the integral can be written as $-\frac{1}{2}\int u^{-\frac{1}{2}}\,du$.

c Hence find the primitive of $\frac{x}{\sqrt{1-x^2}}$.

4 Repeat the previous question for each indefinite integral and substitution.

a $\int x^3(x^4+1)^5\,dx$ (Let $u = x^4 + 1$.)

b $\int x^2\sqrt{x^3-1}\,dx$ (Let $u = x^3 - 1$.)

c $\int \frac{1}{\sqrt{x}(1+\sqrt{x})^3}\,dx$ (Let $u = 1 + \sqrt{x}$.)

d $\int \tan^2 2x \sec^2 2x\,dx$ (Let $u = \tan 2x$.)

5 Find the exact value of each definite integral, using the given substitution.

a $\int_0^1 x^2(2+x^3)^3\,dx$ (Let $u = 2+x^3$.)

b $\int_0^{\frac{\pi}{2}} \cos^2 x \sin x\,dx$ (Let $u = \cos x$.)

c $\int_{\frac{1}{2}\sqrt{3}}^1 x\sqrt{1-x^2}\,dx$ (Let $u = 1-x^2$.)

d $\int_1^{e^2} \frac{\ln x}{x}\,dx$ (Let $u = \ln x$.)

e $\int_0^{\frac{\pi}{4}} \sin^4 2x \cos 2x\,dx$ (Let $u = \sin 2x$.)

f $\int_0^1 \frac{(\sin^{-1} x)^3}{\sqrt{1-x^2}}\,dx$ (Let $u = \sin^{-1} x$.)

DEVELOPMENT

6 Use the substitution $u = x^3$ to find the exact area bounded by the curve $y = \dfrac{x^2}{1+x^6}$, the x-axis and the line $x = 1$.

7 Evaluate each definite integral, using the substitution $u = \sin x$.

a $\int_0^{\frac{\pi}{6}} \frac{\cos x}{1+\sin x}\,dx$

b $\int_0^{\frac{\pi}{2}} \frac{\cos x}{1+\sin^2 x}\,dx$

c $\int_0^{\frac{\pi}{2}} \cos^3 x\,dx$

8 Find each indefinite integral, using the given substitution.

a $\int \frac{e^{2x}}{\sqrt{1+e^{2x}}}\,dx$ (Let $u = e^{2x}$.)

b $\int \frac{1}{x\ln x}\,dx$ (Let $u = \ln x$.)

9 A curve has gradient function $\dfrac{e^{2x}}{1+e^{4x}}$ and passes through the point $\left(0, \frac{\pi}{8}\right)$. Use the substitution $u = e^{2x}$ to find its equation.

10 a Show that $\dfrac{d}{dx}(\sec x) = \sec x \tan x$.

b Use the substitution $u = \sec x$, and the standard form $\displaystyle\int a^x \, dx = \frac{a^x}{\ln a}$, to find:

(i) $\displaystyle\int_0^{\frac{\pi}{3}} 2^{\sec x} \sec x \tan x \, dx$ (ii) $\displaystyle\int_0^{\frac{\pi}{4}} \sec^5 x \tan x \, dx$

11 Evaluate $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\sin 2x}{1+\sin^2 x} \, dx$, using the substitution $u = \sin^2 x$.

12 a Use the substitution $u = \sqrt{x}$ to find $\displaystyle\int \frac{1}{\sqrt{x(1-x)}} \, dx$.

b Evaluate the integral in part a again, using the substitution $u = x - \frac{1}{2}$.

c Hence show that $\sin^{-1}(2x-1) = 2\sin^{-1}\sqrt{x} - \frac{\pi}{2}$, for $0 < x < 1$.

ENRICHMENT

13 Use the substitution $u = x - \dfrac{1}{x}$ to show that $\displaystyle\int_1^{\frac{1}{2}(\sqrt{6}+\sqrt{2})} \frac{1+x^2}{1+x^4}\,dx = \frac{\pi}{4\sqrt{2}}$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **19 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 North Sydney Girls High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q2 — 1 mark

Which integral is obtained when the substitution $u = 1 + 3x$ is applied to $\int x\sqrt{1+3x}\,dx$?

A. $\frac{1}{9}\int (u-1)\sqrt{u}\,du$ B. $\frac{1}{6}\int (u-1)\sqrt{u}\,du$ C. $\frac{1}{3}\int (u-1)\sqrt{u}\,du$ D. $\frac{1}{4}\int (u-1)\sqrt{u}\,du$

Hint: you need both x and dx in terms of u.

PP 2 Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q8 — 1 mark

Which of the following is equivalent to $\int_0^{\frac{\pi}{3}} \cos^5 x \sin x\,dx$, after applying the substitution $u = \cos x$?

(A) $\int_0^{\frac{\pi}{3}} u^5\,du$ (B) $-\int_{\frac{1}{2}}^1 u^5\,du$ (C) $-\int_0^{\frac{\pi}{3}} u^5\,du$ (D) $\int_{\frac{1}{2}}^1 u^5\,du$

Hint: change the limits, then use the minus sign to swap them.

PP 3 Fort Street High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(a) — 2 marks

Find $\int 2x\cos(x^2-1)\,dx$, using $u = x^2 - 1$.

Hint: $du = 2x\,dx$ replaces the whole of $2x\,dx$.

PP 4 James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q13(a) — 2 marks

Use the substitution $u = \sqrt{x}$ to find $\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,dx$.

Hint: $du = \frac{1}{2\sqrt{x}}\,dx$, so $\frac{dx}{\sqrt{x}} = 2\,du$.

PP 5 North Sydney Boys High School, 2023, Year 12 Mathematics Extension 1 Assessment Task 3, Q11(c) — 2 marks

By using substitution $u = \tan^{-1} x$ evaluate $\int \frac{e^{\tan^{-1}x}}{1+x^2}\,dx$.

Hint: $du = \frac{dx}{1+x^2}$.

PP 6 **Blacktown Boys' High School, 2022, HSC Trial Examination, Mathematics Extension 1, Q11(e)** 3 marks

Use the substitution $u = x^3$ to evaluate $\int_0^1 x^2 e^{x^3}\,dx$.

Hint: $x^2\,dx = \frac{1}{3}\,du$; change the limits too.

PP 7 **Kinross Wolaroi School, 2023, Year 12 Mathematics Extension 1 Trial HSC Examination, Q12(a)** 3 marks

Use the substitution $u = \tan x$ to evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \tan^2 x \sec^2 x\,dx$.

Hint: $du = \sec^2 x\,dx$; the new limits are $\tan\frac{\pi}{4}$ and $\tan\frac{\pi}{3}$.

PP 8 **Sydney Grammar School, 2023, Trial Examination, Form VI Mathematics Extension 1, Q12(a)** 3 marks

Calculate $\int_0^1 x(1-x)^7\,dx$ using the substitution $u = 1 - x$.

Hint: like Worked Example 3a: write $x = 1 - u$ as well.

PP 9 **St Aloysius' College, 2024, Trial Higher School Certificate Examination, Mathematics Extension 1, Q12(f)** 2 marks

By letting $u = 2x^3 + 3x$, find the exact value of the integral below:

$$\int_0^1 \frac{2x^2+1}{25+(2x^3+3x)^2}\,dx$$

Hint: $du = (6x^2+3)\,dx = 3(2x^2+1)\,dx$. The result is a $\tan^{-1}$ standard form.

LESSON 12E

Further integration by substitution

Reverse the procedure of Lesson 12D: replace x by a function of u (or of θ), including trigonometric substitutions such as $x = a\sin\theta$.

CONCEPT 1 · substituting x by a function of u

"LET X = SOME FUNCTION OF U"

The second stage of integration by substitution replaces x by a function of u. Then

- find dx in terms of du by differentiating x with respect to u;
- write every part of the integrand in terms of u (for example, if $x = 1 - u^2$ then $\sqrt{1-x} = u$);
- change the limits to values of u, or substitute back at the end for an indefinite integral.

Common choices: $x = u^2 \pm c$ to remove a square root $\sqrt{x \pm c}$; $x = u \pm c$ to simplify a denominator.

TUTOR DEMO **Removing a square root**

Evaluate $\displaystyle\int_0^3 \frac{x}{\sqrt{x+1}}\,dx$, using the substitution $x = u^2 - 1$, where $u > 0$.

YOUR TURN **Same substitution**

Evaluate $\displaystyle\int_0^8 \frac{x}{\sqrt{x+1}}\,dx$, using the substitution $x = u^2 - 1$, where $u > 0$.

WORKED EXAMPLE 1 **Keep for homework**

Find $\displaystyle\int_0^1 x\sqrt{1-x}\,dx$, using the substitution $x = 1 - u^2$.

SOLUTION

Let $x = 1 - u^2$. Then $dx = -2u\,du$, and $\sqrt{1-x} = u$. When $x = 0$, $u = 1$; when $x = 1$, $u = 0$.

$$\begin{aligned}\int_0^1 x\sqrt{1-x}\,dx &= \int_1^0 (1-u^2)\,u\,(-2u)\,du \\ &= -2\int_1^0 (u^2 - u^4)\,du \\ &= -2\left[\tfrac{1}{3}u^3 - \tfrac{1}{5}u^5\right]_1^0 \\ &= -0 + 2\left(\tfrac{1}{3} - \tfrac{1}{5}\right) = \tfrac{4}{15}\end{aligned}$$

This is the integral of Worked Example 3a in Lesson 12D, done a quite different way.

CONCEPT 2 · trigonometric substitutions

X = A SIN Θ, X = A TAN Θ, ...

An expression $\sqrt{a^2 - x^2}$ suggests $x = a\sin\theta$, because then $\sqrt{a^2 - x^2} = \sqrt{a^2\cos^2\theta} = a\cos\theta$. Similarly $a^2 + x^2$ suggests $x = a\tan\theta$, because $a^2 + a^2\tan^2\theta = a^2\sec^2\theta$.

Rule of thumb. Work with positive square roots, and keep θ in the same quadrants as in the definitions of the inverse trigonometric functions (for $x = a\sin\theta$, use $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$, where $\cos\theta \ge 0$).

Strictly, given a value of x there are infinitely many possible θ; choosing θ in this restricted interval removes the ambiguity. A full account is not required in this course.

TUTOR DEMO x = 3 sin θ

Find $\int_0^{\frac{3}{2}} \sqrt{9 - x^2}\,dx$, using the substitution $x = 3\sin\theta$.

YOUR TURN x = 2 sin θ

Find $\int_0^{\sqrt{2}} \sqrt{4 - x^2}\,dx$, using the substitution $x = 2\sin\theta$.

WORKED EXAMPLE 2 Keep for homework

Find $\int_{3\sqrt{2}}^{6} \sqrt{36 - x^2}\,dx$: **a** using the substitution $x = 6\sin u$, **b** using the formula for the area of a segment.

SOLUTION

a Let $x = 6\sin u$. Then $dx = 6\cos u\,du$ and $\sqrt{36 - x^2} = 6\cos u$. When $x = 3\sqrt{2}$, $u = \frac{\pi}{4}$; when $x = 6$, $u = \frac{\pi}{2}$.

$$\begin{aligned}\int_{3\sqrt{2}}^{6} \sqrt{36 - x^2}\,dx &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 6\cos u \times 6\cos u\,du \\ &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 36\left(\tfrac{1}{2} + \tfrac{1}{2}\cos 2u\right)du \\ &= \left[18u + 9\sin 2u\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\ &= (9\pi + 0) - \left(\tfrac{9\pi}{2} + 9\right) \\ &= \tfrac{9}{2}(\pi - 2)\end{aligned}$$

b The integral is the area under the semicircle $y = \sqrt{36 - x^2}$ from $x = 3\sqrt{2}$ to $x = 6$. This is half the segment of the circle of radius 6 cut off by the chord $x = 3\sqrt{2}$, which subtends an angle of $90°$ at the centre. Using segment area $\frac{1}{2}r^2(\theta - \sin\theta)$:

$$\int_{3\sqrt{2}}^{6} \sqrt{36 - x^2}\,dx = \tfrac{1}{2} \times \tfrac{1}{2} \times 6^2\left(\tfrac{\pi}{2} - \sin\tfrac{\pi}{2}\right) = 9\left(\tfrac{\pi}{2} - 1\right)$$

SUMMARY OF LESSONS 12D AND 12E

- The reverse chain rule straightforwardly reverses the steps of differentiation by the chain rule.
- A reverse-chain-rule formula can be developed for each standard form.
- Substitution can be done by "Let u be a function of x".
- Substitution can be done by "Let x be a function of u".
- When substituting into a definite integral, the limits of integration can be substituted as well, so that there is no need ever to return from u to x.

TEXTBOOK HOMEWORK · Exercise 12E

FOUNDATION

1 Consider the integral $I = \int x(x-1)^5 \, dx$, and let $x = u + 1$.

a Show that $dx = du$.

b Show that $I = \int u^5(u+1) \, du$.

c Hence find I.

d Check your answer by differentiating it.

2 Using the same substitution as in the previous question, find:

a $\int \frac{x}{\sqrt{x-1}} \, dx$

b $\int \frac{x}{(x-1)^2} \, dx$

3 Consider the integral $J = \int x\sqrt{x+1} \, dx$, and let $x = u^2 - 1$.

a Show that $dx = 2u \, du$.

b Show that $J = 2\int (u^4 - u^2) \, du$.

c Hence find J.

d Check your answer by differentiating it.

4 Using the same substitution as in the previous question, find:

a $\int x^2\sqrt{x+1} \, dx$

b $\int \frac{2x+3}{\sqrt{x+1}} \, dx$

5 Find each indefinite integral using the given substitution.

a $\int \dfrac{x-2}{x+2}\,dx$ (Let $x = u-2$.)

b $\int \dfrac{2x+1}{\sqrt{2x-1}}\,dx$ (Let $x = \frac{1}{2}(u+1)$.)

c $\int \dfrac{1}{1+\sqrt{x}}\,dx$ (Let $x = (u-1)^2$.)

6 Evaluate, using the given substitution:

a $\int_0^1 x(x+1)^3\,dx$ (Let $x = u-1$.)

b $\int_0^1 \dfrac{3x}{\sqrt{3x+1}}\,dx$ (Let $x = \frac{1}{3}(u-1)$.)

c $\int_0^4 x\sqrt{4-x}\,dx$ (Let $x = 4-u^2$.)

d $\int_0^4 \dfrac{1}{3+\sqrt{x}}\,dx$ (Let $x = (u-3)^2$.)

DEVELOPMENT

7 a Consider the integral $I = \int \dfrac{1}{\sqrt{5-4x-x^2}}\,dx$, and let $x = u-2$. Show that $I = \int \dfrac{1}{\sqrt{9-u^2}}\,du$, and hence find I.

b Use a similar approach to find: (i) $\int \dfrac{1}{x^2+2x+4}\,dx$ (let $x = u-1$) (ii) $\int_1^2 \dfrac{1}{\sqrt{3+2x-x^2}}\,dx$ (let $x = u+1$)

8 a Consider the integral $J = \int \frac{1}{\sqrt{4-x^2}}\,dx$, and let $x = 2\sin\theta$. Show that $J = \int 1\,d\theta$, and hence show that $J = \sin^{-1}\frac{x}{2} + C$.

b Using a similar approach, find: (i) $\int \frac{1}{9+x^2}\,dx$ (let $x = 3\tan\theta$) (ii) $\int \frac{1}{\sqrt{1-4x^2}}\,dx$ (let $x = \frac{1}{2}\sin\theta$) (iii) $\int_0^3 \frac{1}{\sqrt{36-x^2}}\,dx$ (let $x = 6\sin\theta$)

9 a Consider the integral $I = \int \frac{1}{(1-x^2)^{\frac{3}{2}}}\,dx$, and let $x = \sin\theta$. Show that $I = \int \sec^2\theta\,d\theta$, and hence show that $I = \frac{x}{\sqrt{1-x^2}} + C$.

b Similarly, use the given substitution to find: (i) $\int \frac{1}{(4+x^2)^{\frac{3}{2}}}\,dx$ (let $x = 2\tan\theta$) (ii) $\int_0^2 \sqrt{4-x^2}\,dx$ (let $x = 2\sin\theta$)

10 Find the equation of the curve $y = f(x)$ if $f'(x) = \frac{\sqrt{x^2-9}}{x}$ and $f(3) = 0$. (Use the substitution $x = 3\sec\theta$.)

11 Find the exact area of the region bounded by $y = \frac{x^3}{\sqrt{3-x^2}}$, the x-axis and the line $x = 1$. (Use the substitution $x = \sqrt{3}\sin\theta$, followed by the substitution $u = \cos\theta$.)

12 Use integration to confirm that the area of a circle is πr^2. (Find the area bounded by the semicircle $y = \sqrt{r^2 - x^2}$ and the x-axis and double it, using the substitution $x = r\sin\theta$.)

ENRICHMENT

13 a Use the substitution $x = -u$ to show that $\int_{-2}^{2} \frac{x^2}{e^x + 1}\,dx = \int_{-2}^{2} \frac{x^2 e^x}{e^x + 1}\,dx$.

b Hence find $\int_{-2}^{2} \frac{x^2}{e^x + 1}\,dx$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **20 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(b) — 3 marks

Use the substitution $x = 2\sin\theta$ to find $\displaystyle\int_0^1 \sqrt{4-x^2}\,dx$.

Hint: $\sqrt{4-x^2} = 2\cos\theta$; when $x = 1$, $\sin\theta = \frac{1}{2}$.

PP 2 Gosford High School, 2022, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(d) — 3 marks

Using the substitution $u^2 = x+1$, where $u > 0$, find $\displaystyle\int_0^3 \frac{x+2}{\sqrt{x+1}}\,dx$.

Hint: $x = u^2 - 1$, so $dx = 2u\,du$ and $x + 2 = u^2 + 1$.

PP 3 Hornsby Girls High School, 2022, Year 12 Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(d) — 3 marks

Use the substitution $x = u^2 - 1\ (u > 0)$ to evaluate $\displaystyle\int_0^3 x\sqrt{x+1}\,dx$.

Hint: this is textbook question 3 as a definite integral.

PP 4 Normanhurst Boys' High School, 2023, Year 12 Mathematics Extension 1 Course Assessment Task 4 (Trial Examination), Q12(c) — 3 marks

By using the substitution $x = \frac{1}{4}(u^2+5)$, find $\displaystyle\int 3x\sqrt{4x-5}\,dx$.

Hint: $4x - 5 = u^2$, and $dx = \frac{1}{2}u\,du$. Substitute back at the end.

PP 5 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q13(d)** 4 marks

Find the exact value of $\int_0^{\frac{1}{2}} \dfrac{x^2}{\sqrt{1-x^2}}\,dx$, using the substitution $x = \sin\theta$.

Hint: the integrand becomes $\sin^2\theta$; use Lesson 12C.

PP 6 **Ascham School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q11(d)** 4 marks

(i) Show that $\dfrac{u^2}{1+u^2} = 1 - \dfrac{1}{1+u^2}$. 1

(ii) Using the substitution $x = u^2$, $u \geq 0$, and the result of part (i), find $\displaystyle\int \frac{\sqrt{x}}{1+x}\,dx$. 3

Hint: $dx = 2u\,du$ and $\sqrt{x} = u$; part (i) turns the integrand into a standard form.

LESSON 12F

Volumes of rotation

Find the volume of a solid of revolution formed by rotating a region about the x-axis or the y-axis, including regions between two curves.

CONCEPT 1 · rotating a region about the x-axis

WHERE THE FORMULA COMES FROM

When a region in the plane is rotated about the x-axis or the y-axis, a solid of revolution is formed, like shaping wood on a lathe. Take the region under $y = f(x)$ for $a \leq x \leq b$ and rotate it about the x-axis.

Imagine the solid sliced perpendicular to the x-axis into infinitely many circular slices, each of width dx. The vertical strip of height y generates one slice:

- radius of circular slice $= y$, the height of the strip,
- area of circular slice $= \pi y^2$, because it is a circle,
- the slice is a thin cylinder of thickness dx, so its volume is $\pi y^2\,dx$ (area × thickness).

Adding all the slices from $x = a$ to $x = b$ gives the total volume.

VOLUMES OF REVOLUTION ABOUT THE X-AXIS

The volume of the solid generated when the region between a curve and the x-axis from $x = a$ to $x = b$ is rotated about the x-axis is

$$V = \int_a^b \pi y^2\,dx \text{ cubic units.}$$

If the curve is below the x-axis, the volume is still positive, because y^2 rather than y occurs in the formula. Unless units are given, use cubic units (u³).

TUTOR DEMO **About the x-axis**

The region under the semicircle $y = \sqrt{9 - x^2}$ from $x = 1$ to $x = 3$ is rotated about the x-axis. Find the volume generated.

YOUR TURN **Same method**

The region under $y = \sqrt{x}$ from $x = 0$ to $x = 4$ is rotated about the x-axis. Find the volume generated.

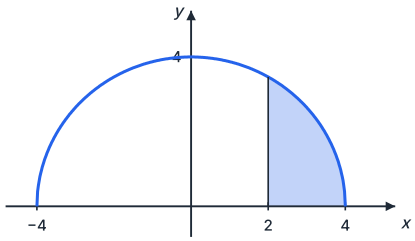
WORKED EXAMPLE 1 **Keep for homework**

The shaded region cut off the semicircle $y = \sqrt{16 - x^2}$ by the line $x = 2$ is rotated about the x-axis. Find the volume generated.

SOLUTION

Squaring, $y^2 = 16 - x^2$, so

$$\begin{aligned} V &= \int_2^4 \pi y^2 \, dx \\ &= \pi \int_2^4 (16 - x^2) dx \\ &= \pi \left[16x - \tfrac{1}{3}x^3\right]_2^4 \\ &= \pi \left(64 - \tfrac{64}{3} - 32 + \tfrac{8}{3}\right) \\ &= \tfrac{40\pi}{3} \text{ cubic units.} \end{aligned}$$



CONCEPT 2 · volumes of revolution about the y-axis

VOLUMES OF REVOLUTION ABOUT THE Y-AXIS

To rotate about the y-axis, exchange x and y. The volume of the solid generated when the region between a curve and the y-axis from $y = a$ to $y = b$ is rotated about the y-axis is

$$V = \int_a^b \pi x^2 \, dy \text{ cubic units.}$$

When y is given as a function of x, rewrite the equation with x^2 as the subject, and use y-values for the limits.

TUTOR DEMO **About the y-axis**

The region between the curve $y = x^2$, the y-axis, and the lines $y = 1$ and $y = 4$, in the first quadrant, is rotated about the y-axis. Find the volume of the solid.

YOUR TURN **Same method**

The region between the curve $x = y^2$, the y-axis and the line $y = 2$ is rotated about the y-axis. Find the volume of the solid.

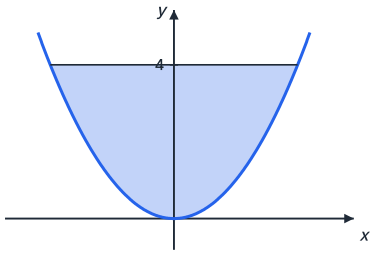
WORKED EXAMPLE 2 **Keep for homework**

Find the volume of the solid formed by rotating the region between $y = x^2$ and the line $y = 4$ about the y-axis.

SOLUTION

Because $x^2 = y$,

$$\begin{aligned} V &= \int_0^4 \pi x^2 \, dy \\ &= \pi \int_0^4 y \, dy \\ &= \pi \left[\tfrac{1}{2} y^2 \right]_0^4 \\ &= 8\pi \text{ cubic units.} \end{aligned}$$



CONCEPT 3 · volume by subtraction

ROTATING THE REGION BETWEEN TWO CURVES

When rotating the region between two curves lying above the x-axis, the two integrals are subtracted, as if the outer volume had been formed first and then the inner volume cut away from it. If the two integrals have the same limits, combine them:

$$V = \int_a^b \pi \left(y_2^2 - y_1^2 \right) dx \quad (y_2 > y_1 > 0) \qquad V = \int_a^b \pi \left(x_2^2 - x_1^2 \right) dy \quad (x_2 > x_1 > 0)$$

Note: $\pi(y_2^2 - y_1^2)$ is **not** $\pi(y_2 - y_1)^2$. Also, the volumes about the two axes are not normally equal: an element of area generates a larger element of volume when it is further from the axis of rotation.

TUTOR DEMO **Between a line and a parabola**

The region between $y = x$ and $y = x^2$ is rotated about the x-axis. Find the volume generated.

YOUR TURN **The same region, other axis**

The same region, between $y = x$ and $y = x^2$, is rotated about the y-axis. Find the volume generated.

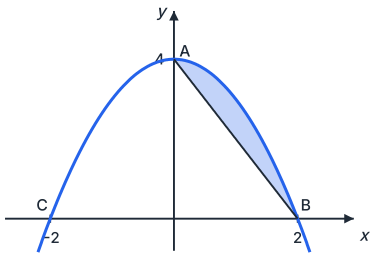
WORKED EXAMPLE 3 **Keep for homework**

The curve $y = 4 - x^2$ meets the y-axis at $A(0, 4)$ and the x-axis at $B(2, 0)$ and $C(-2, 0)$. Find the volume generated when the region between the curve and the line AB is rotated **a** about the x-axis, **b** about the y-axis.

SOLUTION

a The parabola is $y_2 = 4 - x^2$, and the line AB is $y_1 = 4 - 2x$.

$$\begin{aligned} V &= \int_0^2 \pi \left(y_2^2 - y_1^2\right) dx \\ &= \pi \int_0^2 \Big((16 - 8x^2 + x^4) - (16 - 16x + 4x^2) \Big) dx \\ &= \pi \int_0^2 \left(x^4 - 12x^2 + 16x\right) dx \\ &= \pi \Big[\tfrac{1}{5}x^5 - 4x^3 + 8x^2 \Big]_0^2 \\ &= \pi \left(\tfrac{32}{5} - 32 + 32 \right) = \tfrac{32\pi}{5} \text{ u}^3 \end{aligned}$$



b The parabola is $x_2^2 = 4 - y$, and the line is $x_1 = 2 - \frac{1}{2}y$.

$$\begin{aligned} V &= \int_0^4 \pi \left(x_2^2 - x_1^2\right) dy \\ &= \pi \int_0^4 \Big((4 - y) - \left(4 - 2y + \tfrac{1}{4}y^2\right) \Big) dy \\ &= \pi \int_0^4 \left(-\tfrac{1}{4}y^2 + y\right) dy \\ &= \pi \Big[-\tfrac{1}{12}y^3 + \tfrac{1}{2}y^2 \Big]_0^4 \\ &= \pi \left(-\tfrac{16}{3} + 8\right) = \tfrac{8\pi}{3} \text{ u}^3 \end{aligned}$$

CONCEPT 4 · cones and spheres

VOLUME OF CONES AND SPHERES

For a cone, $V = \frac{1}{3}\pi r^2 h$. For a sphere, $V = \frac{4}{3}\pi r^3$.

These formulae can only be proven using integration: rotate the line $y = \frac{r}{h}x$ (cone) or the semicircle $y = \sqrt{r^2 - x^2}$ (sphere) about the x-axis. Homework question 13 works through the proofs.

TEXTBOOK HOMEWORK · Exercise 12F

FOUNDATION

1 a Sketch the region bounded by the line $y = 3x$ and the x-axis between $x = 0$ and $x = 3$.

b When this region is rotated about the x-axis, a right circular cone is formed. Find the radius and height of the cone and hence find its volume.

c Evaluate $\pi \int_0^3 y^2 \, dx = \pi \int_0^3 9x^2 \, dx$ in order to check your answer.

2 a Sketch the region bounded by the curve $y = \sqrt{9 - x^2}$ and the x-axis between $x = -3$ and $x = 3$.

b When this region is rotated about the x-axis, a sphere is formed. Find the radius of the sphere and hence find its volume.

c Evaluate $\pi \int_{-3}^3 y^2 \, dx = \pi \int_{-3}^3 (9 - x^2) \, dx$ in order to check your answer.

3 Calculate the volume generated when each region is rotated about the x-axis.

a The region between the line $y = 2$ and the x-axis, from $x = -2$ to $x = 2$.

b The region under $y = x^2$, from $x = 0$ to $x = 2$.

c The region under $y = \sqrt{x}$, from $x = 2$ to $x = 4$.

d The region under $y = \sqrt{4 - x^2}$, from $x = 0$ to $x = 2$.

e The region between the line $y = x + 2$ and the x-axis, from $x = -5$ to $x = -2$.

4 Calculate the volume generated when each region is rotated about the y-axis.

a The region between the line $x = 2y$ and the y-axis, from $y = 0$ to $y = 4$.

b The region between the curve $x = y^2$ and the y-axis, from $y = 2$ to $y = 5$.

c The region between the curve $x = \sqrt{16 - y^2}$ and the y-axis, from $y = -4$ to $y = 4$.

d The region between the curve $x = -y^2$ and the y-axis, from $y = -3$ to $y = 0$.

DEVELOPMENT

5 The region between the curve $y = e^x$ and the x-axis from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume of the solid generated.

6 Find the volume generated when the region between the curve $y = \dfrac{1}{\sqrt{x}}$ and the x-axis, from $x = 2$ to $x = 4$, is rotated about the x-axis.

7 The region between the curve $y = \dfrac{1}{x^2}$ and the y-axis from $y = 1$ to $y = 6$ is rotated about the y-axis. Calculate the exact volume of the solid formed.

8 a Write $\tan^2 x$ in terms of $\sec^2 x$.

b The region bounded by the curve $y = \tan x$, the x-axis and the vertical line $x = \frac{\pi}{3}$ is rotated about the x-axis. Find the volume of the solid generated.

9 a Write $\sin^2 x$ in terms of $\cos 2x$.

b The region bounded by the curve $y = \sin x$, the x-axis and the vertical line $x = \frac{\pi}{2}$ is rotated about the x-axis. Find the volume of the solid generated.

10 Find the volume of the solid formed by rotating the region with the given boundaries about the x-axis. (A sketch of each region will be needed.)

a $y = x + 3$, $x = 3$, $x = 5$ and $y = 0$

b $y = 1 + \sqrt{x}$, $x = 1$, $x = 4$ and $y = 0$

c $y = x^3 - x$ and $y = 0$

11 Find the volume of the solid formed by rotating the region with the given boundaries about the y-axis. (A sketch of each region will be needed.)

a $x = y - 2$, $y = 1$ and $x = 0$

b $x = y^2 + 1$, $y = 0$, $y = 1$ and $x = 0$

c $y = 1 - x^2$ and $y = 0$

12 The region R is bounded by the parabola $y = x^2$ and the x-axis from $x = 0$ to $x = 4$.

a Find the volume of the cylinder formed when the region between the line $x = 4$ and the y-axis from $y = 0$ to $y = 16$ is rotated about the y-axis.

b Find the volume of the solid formed when the region between the parabola $y = x^2$ and the y-axis from $y = 0$ to $y = 16$ is rotated about the y-axis.

c Hence find the volume of the solid formed when R is rotated about the y-axis.

13 In this question some standard volume formulae will be proven.

a A right circular cone of height h and radius r is generated by rotating about the x-axis the region bounded by the line $y = \dfrac{rx}{h}$, the x-axis and the line $x = h$. Show that the volume of the cone is $\frac{1}{3}\pi r^2 h$.

b A cylinder of height h and radius r is generated by rotating about the x-axis the region bounded by the line $y = r$, the x-axis, the y-axis and the line $x = h$. Show that the volume of the cylinder is $\pi r^2 h$.

c (i) A sphere of radius r is generated by rotating about the x-axis the region between the semicircle $y = \sqrt{r^2 - x^2}$ and the x-axis. Show that the volume of the sphere is $\frac{4}{3}\pi r^3$. (ii) A spherical cap of height h is formed by rotating about the x-axis the region between the semicircle $y = \sqrt{r^2 - x^2}$ and the x-axis from $x = r - h$ to $x = r$. Show that the volume of the cap is $\frac{1}{3}\pi h^2(3r - h)$.

14 **a** Sketch the region bounded by the curves $y = x^2$ and $y = x^3$.

b Find the volume of the solid generated when this region is rotated about: (i) the x-axis, (ii) the y-axis.

15 **a** On the same number plane sketch the graphs of $xy = 5$ and $x + y = 6$, clearly indicating their points of intersection.

b Find the volume of the solid generated when the region bounded by the two curves is rotated about the x-axis.

ENRICHMENT

16 Consider the curves $f(x) = x^n$ and $f(x) = x^{n+1}$, where n is a positive integer.

a Find the points of intersection of the curves.

b Show that the volume V_n of the solid generated when the region bounded by the two curves is rotated around the x-axis is given by $\pi\left(\dfrac{1}{2n+1} - \dfrac{1}{2n+3}\right)$ cubic units.

c Describe the solid whose volume is given by $\lim\limits_{n\to\infty}(V_1 + V_2 + V_3 + \cdots + V_n)$.

d Find the volume of the solid in part c.

e Deduce that the series $\dfrac{1}{3\times 5} + \dfrac{1}{5\times 7} + \dfrac{1}{7\times 9} + \cdots$ has a limiting sum of $\dfrac{1}{6}$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **20 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 Gosford High School, 2022, Trial Higher School Certificate Examination, Mathematics Extension 1, Q5 — 1 mark

Which integral finds the volume of the solid formed when $y = \sin x$ is rotated around the x-axis between $x = 0$ and $x = \frac{\pi}{3}$?

A. $\frac{\pi}{2}\int_0^{\frac{\pi}{3}} \cos 2x - 1\,dx$

B. $\frac{\pi}{2}\int_0^{\frac{\pi}{3}} 1 - \cos 2x\,dx$

C. $\frac{\pi}{2}\int_0^{\frac{\pi}{3}} 1 + \cos 2x\,dx$

D. $\pi\int_0^{\frac{\pi}{3}} \cos 2x - 1\,dx$

Hint: $V = \pi\int y^2\,dx$ and $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$.

PP 2 Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q2 — 1 mark

The curve $y = \sqrt{16 - (x+1)^2}$ is rotated about the x-axis to form a solid of revolution. Which of the following integrals would give the volume of the solid?

A. $\pi\int_{-4}^{4} \left(16 - (x+1)^2\right)dx$

B. $\pi\int_{-5}^{3} \left(\sqrt{16 - (x+1)^2}\right)dx$

C. $\pi\int_{-5}^{3} \left(16 - (x+1)^2\right)^2 dx$

D. $\pi\int_{-5}^{3} \left(16 - (x+1)^2\right)dx$

Hint: the curve is a semicircle; find where it meets the x-axis for the limits.

PP 3 Sydney Grammar School, 2021, Trial Examination, Form VI Mathematics Extension 1, Q12(a) — 3 marks

The region enclosed by the graph of $y = \frac{1}{\sqrt{x}}$, the vertical lines $x = 2$ and $x = 8$, and the x-axis is rotated about the x-axis. Find the exact volume of the solid created. Express your answer in simplified form.

Hint: $y^2 = \frac{1}{x}$. Simplify the logarithm at the end.

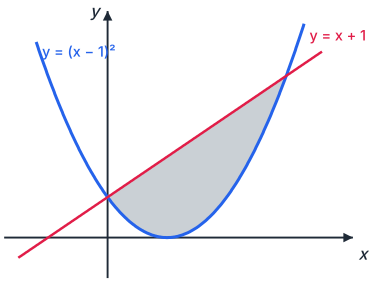
PP 4 Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(b) — 3 marks

The region bounded by the curve $y = \ln x$, the x-axis, the y-axis and the line $y = \ln 4$ is rotated about the y-axis. Find the exact volume of the solid of revolution.

Hint: rotating about the y-axis needs x^2 in terms of y: $x = e^y$.

PP 5 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(d)** 4 marks

The shaded area is bounded by $y = (x-1)^2$ and $y = x + 1$. The area is rotated around the x-axis to form a solid. Find the volume of the solid generated.



Hint: find the intersections, then use $\pi \int (y_2^2 - y_1^2)\, dx$.

PP 6 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q11(c)** 3 marks

The region bounded by the function $y = \dfrac{2}{\sqrt{x^2+3}}$, the x-axis, the y-axis and the line $x = 1$, is shaded. Find the exact volume when this region is rotated about the x-axis.

Hint: $y^2 = \dfrac{4}{x^2+3}$, which is a $\tan^{-1}$ standard form from Lesson 12B.

PP 7 **Ascham School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q11(c)** 5 marks

(i) Sketch the graph of $y = 2\sin^{-1} x$ showing clearly the coordinates of the endpoints. 2

(ii) The area in the first quadrant bounded by the curve $y = 2\sin^{-1} x$, the y-axis and the line $y = \dfrac{\pi}{2}$ is rotated about the y-axis. Find the volume of the resulting solid. 3

Hint: make x the subject: $x = \sin \frac{y}{2}$. Then use Lesson 12C for $\int \sin^2 \frac{y}{2}\, dy$.

CHAPTER 12 REVIEW

Chapter review exercise

Mixed practice from the whole chapter.

1 Find the derivative of:

a $y = \sin^{-1} 3x$ b $y = \tan^{-1} \frac{x}{3}$ c $y = \cos^{-1}(1 - x)$

d $y = x^2 \tan^{-1} x$ e $y = \tan^{-1}\left(\frac{1}{2}x + 1\right)$ f $y = \sin^{-1} \frac{1}{x}$

2 Show that $y = \tan^{-1} x$ satisfies the differential equation $(1 + x^2)y'' + 2xy' = 0$.

3 a Show that the functions $y = \cos^{-1} x$ and $y = \sin^{-1} \sqrt{1 - x^2}$ have the same derivative for $0 \leq x \leq 1$.

b Explain the significance of the result in part a.

4 Find:

a $\int \frac{1}{3 + x^2}\, dx$ b $\int \frac{1}{\sqrt{3 - x^2}}\, dx$

c $\int \frac{1}{9 + 4x^2}\, dx$ d $\int \frac{-1}{\sqrt{16 - 9x^2}}\, dx$

5 Evaluate:

a $\int_{\frac{1}{3}}^{\frac{1}{\sqrt{3}}} \frac{1}{1 + 9x^2}\, dx$ b $\int_{-\frac{3}{4}}^{\frac{3}{4}} \frac{1}{\sqrt{3 - 4x^2}}\, dx$

6 Find:

a $\int \cos^2 x \, dx$

b $\int \sin^2 x \, dx$

c $\int \cos^2 2x \, dx$

d $\int \sin^2 4x \, dx$

7 Find the exact value of:

a $\int_0^{\frac{\pi}{3}} \sin^2 3x \, dx$

b $\int_0^{\frac{\pi}{6}} \cos^2 \frac{1}{2}x \, dx$

8 Show that $\int_0^{\frac{\pi}{2}} \cos^2 x \, dx = \int_0^{\frac{\pi}{2}} \cos^2 2x \, dx = \int_0^{\frac{\pi}{2}} \cos^2 4x \, dx = \frac{\pi}{4}$.

9 Find each indefinite integral using the given substitution.

a $\int 5(5x-1)^5 \, dx$ [Let $u = 5x - 1$.]

b $\int 2x(x^2+2)^2 \, dx$ [Let $u = x^2 + 2$.]

c $\int \frac{4x^3}{(x^4+1)^2} \, dx$ [Let $u = x^4 + 1$.]

d $\int \frac{1}{\sqrt{4x+3}} \, dx$ [Let $u = 4x + 3$.]

e $\int \sin^2 x \cos x \, dx$ [Let $u = \sin x$.]

f $\int \tan^3 x \sec^2 x \, dx$ [Let $u = \tan x$.]

10 Evaluate each definite integral using the given substitution.

a $\int_{-1}^{0} x^2(1+x^3)^4\,dx$ [Let $u = 1+x^3$.]

b $\int_{0}^{\frac{\pi}{2}} \cos^3 x \sin x\,dx$ [Let $u = \cos x$.]

c $\int_{1}^{\sqrt{2}} x\sqrt{x^2-1}\,dx$ [Let $u = x^2-1$.]

d $\int_{1}^{e} \frac{(\ln x)^2}{x}\,dx$ [Let $u = \ln x$.]

e $\int_{\frac{1}{2}}^{1} \frac{e^{\frac{1}{x}}}{x^2}\,dx$ [Let $u = \frac{1}{x}$.]

f $\int_{0}^{\frac{\pi}{8}} \frac{\sec^2 2x}{1+\tan 2x}\,dx$ [Let $u = \tan 2x$.]

11 Find each indefinite integral using the given substitution.

a $\int \frac{x}{x-1}\,dx$ [Let $x = u+1$.]

b $\int \frac{x-1}{\sqrt{x+2}}\,dx$ [Let $x = u-2$.]

c $\int x\sqrt{2x+1}\,dx$ [Let $x = \frac{1}{2}u^2 - \frac{1}{2}$.]

d $\int \frac{1}{4+\sqrt{x}}\,dx$ [Let $x = (u-4)^2$.]

12 Evaluate, using the given substitution:

a $\int_{1}^{2} x(x-1)^4\,dx$ [Let $x = u+1$.]

b $\int_{1}^{5} \frac{x}{x+3}\,dx$ [Let $x = u-3$.]

c $\int_{0}^{15} \frac{x}{\sqrt{x+1}}\,dx$ [Let $x = u^2-1$.]

d $\int_{2}^{3} \frac{1}{2}x\sqrt{x-2}\,dx$ [Let $x = u^2+2$.]

13 a State the domain and range of the function $y = \sqrt{9 - x}$.

b Sketch the graph of the function.

c Calculate the area of the region bounded by the curve and the coordinate axes.

d Calculate the volume of the solid formed when this region is rotated about: (i) the x-axis, (ii) the y-axis.

14 A horn is created by rotating about the x-axis the region between the curve $y = \dfrac{1}{\sqrt{4 - x}}$ and the x-axis from $x = 0$ to $x = 3\frac{3}{4}$. Find the volume of the horn.

15 A vase is designed by rotating the parabola $y^2 = 18(x - 6)$ from $y = -6$ to $y = 6$ about the y-axis. Find the exact volume of the vase.

16 a Write $\cos^2 2x$ in terms of $\cos 4x$.

b The region bounded by the curve $y = \cos 2x$ and the x-axis from $x = -\frac{\pi}{6}$ to $x = \frac{\pi}{6}$ is rotated about the x-axis. Show that the solid generated has volume $\dfrac{\pi}{24}\left(4\pi + 3\sqrt{3}\right)$ u³.

17 A tank is created by rotating about the x-axis the region between the curve $y = 1 + e^{-x}$ and the x-axis from $x = 1$ to $x = 3$. Find its volume correct to two decimal places.

18 a Sketch $y = 1 - \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, and shade the region R bounded by the curve and the coordinate axes.

b Find the volume generated when R is rotated about the x-axis.

19 a Find the stationary points of the curve $y = x + \dfrac{1}{x}$ and sketch its graph.

b Show that the line $y = \frac{5}{2}$ intersects the curve when $x = \frac{1}{2}$ or $x = 2$.

c Find the volume of the solid generated when the region between the curve and the line $y = \frac{5}{2}$ is rotated about the x-axis.

20 a On the same set of axes sketch the curves $y = \sqrt{9 - x^2}$ and $y = 18 - 2x^2$.

b Find the area bounded by the two curves.

c Find the volume of the solid formed when this region is rotated about the x-axis.

21 The region under the graph $y = 2^{x+1}$ between $x = 1$ and $x = 3$ is rotated about the x-axis.

a Write down the definite integral representing the volume of the solid that is formed.

b Use the trapezoidal rule with five function values to approximate the volume of the solid, giving your answer as a multiple of π.

c Find the exact value of the volume, then approximate it as an integer multiple of π. Why is the exact answer smaller than the trapezoidal-rule approximation?